

REVIEW

Open Access



Green metal–organic frameworks for circular soil remediation through pollutant sequestration and nutrient regeneration

Irfan Haidri¹ , Aneeza Ishfaq² , Athakorn Promwee¹ and Faisal Mahmood^{2*}

*Correspondence:

Faisal Mahmood

faisalmahmood@gcuf.edu.pk

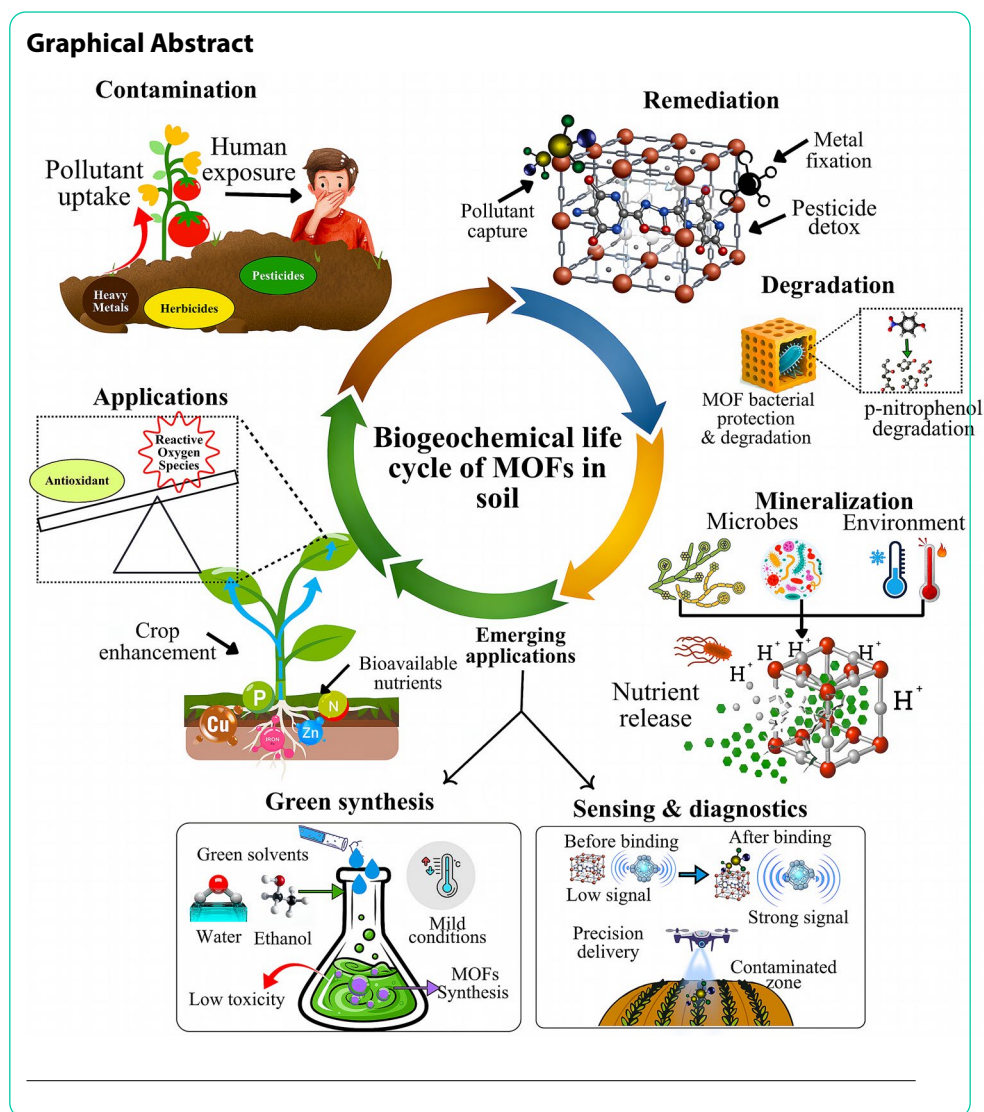
¹School of Agricultural Technology
and Food Industry, Walailak
University,
Nakhon Si Thammarat
80160, Thailand

²Department of Environmental
Sciences, Government College
University Faisalabad,
Faisalabad 38040, Pakistan

Abstract

Agricultural soils are increasingly threatened by contamination from pesticides, herbicides, and potentially toxic metals, posing significant risks to food security, ecosystem functioning, and long-term agricultural sustainability. Conventional remediation strategies primarily focus on contaminant removal but often fail to promote resource recovery and soil regeneration. Metal–organic frameworks (MOFs) have emerged as versatile materials capable of addressing both environmental remediation and nutrient management challenges within a green and circular chemistry framework. Recent advances demonstrate that MOFs can achieve contaminant removal efficiencies exceeding 90% and pesticide loading capacities of up to 90.79%, highlighting their potential as highly effective remediation materials. Beyond pollutant sequestration, MOFs undergo microbial and environmental transformations that facilitate the gradual release of plant-essential nutrients, including phosphorus and iron, thereby enhancing nutrient-use efficiency, supporting nutrient recycling, and improving plant performance. Unlike previous reviews that primarily discuss MOFs either as adsorbents for environmental remediation or as carriers for agrochemical delivery, this review presents an integrated perspective on MOFs as regenerative materials that connect pollutant sequestration with nutrient regeneration through biologically mediated transformation. It brings together recent developments in the field of MOF–microbiome interactions, microbial degradation pathways, nutrient recycling, and circular soil remediation in a single conceptual framework. Furthermore, this review identifies critical knowledge gaps related to field-scale validation, standardized environmental risk assessment, scalable green synthesis, economic feasibility, and the long-term fate of MOF transformation products. By integrating these emerging research directions, this review may provides a roadmap for developing next-generation MOF technologies for resilient and circular agricultural systems.

Keywords Metal-organic frameworks, Soil remediation, Contaminant sequestration, Biogeochemical cycle, Mineralization, Microbial interaction



1 Introduction

Agricultural soils worldwide are increasingly loaded by the long-term accumulation of persistent contaminants resulting from decades of intensive agrochemical use [1, 2]. The use of pesticides, herbicides and inorganic amendments has led to the accumulation of substances that are not biodegradable, such as organochlorine compounds, triazine herbicides and toxic metals such as cadmium (Cd), lead (Pb) and arsenic (As) [3, 4]. These pollutants are not confined to the upper surface of the soil, but they move through the soil pore systems, dissolve into the soil water and ultimately enter the groundwater systems or accumulate in crops [5, 6]. In polluted paddy soils, the level of As can reach 50 mg kg^{-1} , which is about 250 times higher than the generally accepted safety limit of 20 mg kg^{-1} , posing a significant risk of entering the food chain and long-term human exposure [7, 8].

The traditional remediation methods remain limited in their effectiveness, as they often reduce contaminant concentrations at the loss of soil structural integrity and biological functionality [9]. Soil excavation, chemical washing, and bulk amendment with

lime and sulfur are used to reduce contaminant levels, but they usually disturb the natural structure and biological activity of soils [10, 11]. Such methods affect bulk density and soil porosity by destabilizing soil aggregates and affecting beneficial microbial communities that regulate nutrient cycling and plant growth [12]. Conventional sorbents like activated carbon and biochar are also used extensively for contaminant removal, but their adsorption performance and selectivity depend on the contaminant and environmental conditions. These materials can effectively immobilize many contaminants, but can also under certain conditions adsorb certain beneficial nutrients, which can affect nutrient availability in soil [13]. Nanotechnology-based materials have also shown considerable potential for improving soil remediation efficiency. However, further research is still required to better understand their long-term environmental fate and ecological impacts under field conditions [14–17].

In this context, MOFs have emerged as a promising class of green functional materials that combine high remediation performance with environmental sustainability and resource circularity [18]. Consistent with the principles of green chemistry, MOFs are designed not only to capture and immobilize contaminants but also to undergo controlled transformation into beneficial soil constituents, minimizing secondary pollution and waste generation. Their tunable structure, high surface area, and multifunctional properties enable efficient pollutant sequestration while supporting sustainable soil restoration. MOFs have been shown to be effective remediators in certain laboratory and greenhouse experiments. For instance, Fe-based MOFs, UiO-66 derivatives, and MOF–biochar composites have been shown to achieve contaminant removal efficiencies greater than 90% in controlled aqueous environments and artificially contaminated soils, though the effectiveness depends on the chemistry of the contaminant, soil properties, pH, and competing ions [19, 20]. Simultaneously, these materials can stabilize useful microorganisms under stressful conditions, and the activity of microorganisms can be preserved more during a long period of time in nutrient-limited or toxic environments [21, 22]. This dual functionality enables MOFs to eliminate contaminants and facilitate biological remediation at the same time. Unlike persistent nanomaterials, MOFs are synthesized using biologically metal nodes, such as iron (Fe), zinc (Zn), magnesium (Mg), and calcium (Ca), and metabolizable organic linkers, which are designed to be degraded by natural soil processes [23, 24]. The framework structure breaks down and releases these elements gradually into the soil and plant-available form [25]. Nutrient uptake improvements of approximately 57–64.4% have been reported primarily for phosphorus and Mg released from Fe- and Mg-based MOFs in greenhouse experiments with crops such as maize and faba bean grown under nutrient-deficient conditions [26, 27]. This shows the MOFs contribution to soil fertility as a remediation agent and a source of nutrients simultaneously [28].

Although numerous reviews have been published on MOFs for environmental remediation, pollutant adsorption, controlled agrochemical delivery and sustainable agriculture, most of these reviews focus on one application at a time. There is a lack of understanding regarding the combined role of MOFs in pollutant sequestration, nutrient regeneration by biological processes, microbial interactions and circular soil restoration. This review addresses this knowledge gap by providing a unified perspective on the complete life cycle of MOFs in agricultural soils, from contaminant capture and microbial transformation to nutrient recycling and ecosystem restoration. In addition, it

highlights the existing challenges in the areas of environmental safety, field-scale validation, scalable green synthesis, and long-term environmental fate, thus providing possible future research directions for the development of sustainable MOF-based technologies for circular agriculture.

2 Tailored sorption and sequestration dynamics

The application of MOFs as soil remediation agents is based on their high porosity and the chemical programmability of their inner surfaces. In contrast to conventional soil amendments, MOFs can be designed to be highly selective, targeting hazardous legacy pollutants and important soil constituents. The synergistic action of the topology of the framework and functionalization of its surface controls this selective sequestration [29].

2.1 Selective capture of organic micropollutants

Agricultural soils are usually difficult to remediate because several pollutants can occur together, such as persistent organochlorines, modern herbicides, and other organic micropollutants [30, 31]. Such compounds have a high affinity for soil particles and cannot be degraded by natural processes [32]. MOFs offer a structurally tunable material capable of capturing these contaminants by molecular recognition and entrapment within well-defined pore structures [33]. Adsorption of aromatic pesticides is primarily done by non-covalent interactions between the π -electron systems of the framework linkers and the aromatic rings of the pesticide molecules, as illustrated in Fig. 1 [34]. As an example, UiO-66 and UiO-66-NH₂ with an amino-functional group allow π - π stacking and hydrophobic interactions to increase the capture of legacy pesticides. The hydrophobic pyrethroid pesticide Lambda-cyhalothrin is highly interactive with the aromatic cavity of zirconium-based structures and achieves an encapsulation efficiency of 87.71% [35]. Adsorption performance is commonly interpreted using the Langmuir and Freundlich isotherm models, while adsorption kinetics are typically described by pseudo-first-order and pseudo-second-order models. These models provide valuable insights into adsorption behavior and facilitate comparisons among different MOF systems.

The adsorption of relatively polar pesticides is further enhanced by the introduction of functional groups to organic linkers, e.g., aminoterephthalic acid [36]. These groups permit further hydrogen bonding and dipole-dipole interaction with pesticide molecules. For example, amino-modified UiO-66 exhibits an acetamiprid loading capacity of 90.79% in pesticide-delivery studies. This value reflects the strong affinity of the MOF toward the pesticide molecule and demonstrates its adsorption capability [19]. Besides adsorption via weak intermolecular forces, iron-based MOFs, including MIL-101(Fe), may be involved in surface redox reactions that facilitate the conversion or stabilization of reactive organic pollutants. The specific surface area of these materials is often more than 2000 m² · g⁻¹, which gives a high concentration of active adsorption sites to pesticides like Azoxystrobin (AZOX) as shown in Fig. 1 (C) [37].

In natural soil, competitive adsorption by the soil organic matter (SOM) may restrain the removal of the pollutants since the SOM molecules have the potential to occupy the same binding sites as the target contaminants [38, 39]. Nevertheless, the rigid and precisely defined pore openings of MOFs serve as molecular sieves. Small micropollutants, which are usually less than 1.5 nm, are allowed to enter the internal pore system, but large organic molecules like humic and fulvic acids are not as illustrated in Fig. 2. This

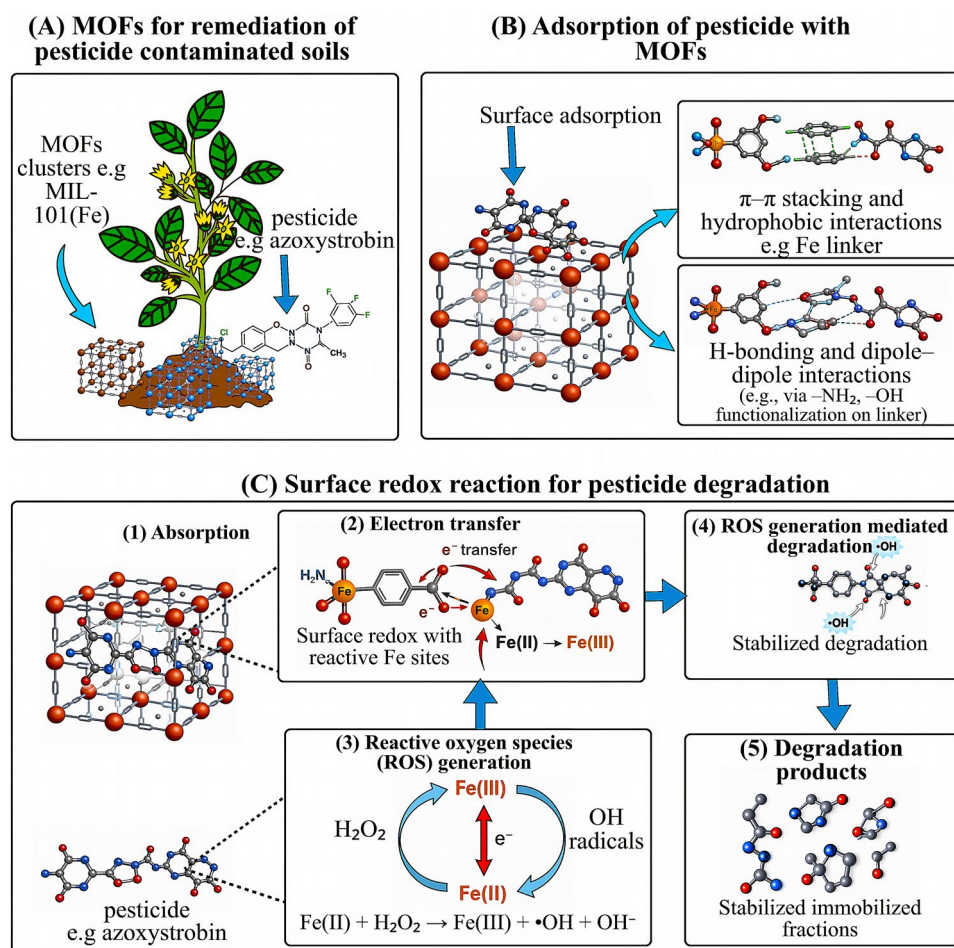


Fig. 1 Conceptual overview of MOF-mediated remediation of pesticide-contaminated soils. The schematic illustration of **(A)** adsorption, **(B)** surface redox reactions, and **(C)** The formation of reactive oxygen species that cause the degradation and stabilization of pesticides in MOF structures

size-selective mechanism maintains the internal adsorption surface of toxic substances even in high-SOM soils [17].

2.2 Biochar-MOF nanocomposites for enhanced sorption

Combining MOFs with biochar is a good approach to soil remediation. Pure MOFs are highly porous with well-defined chemical structures; however, they tend to be mechanically weak and costly to use at a large scale [40, 41]. Conversely, biochar is cheap and mechanically stable, but is not very selective to certain contaminants. The direct growth of MOF crystals within the carbon structure of biochar will combine the benefits of both materials and minimize their respective drawbacks, as demonstrated in Fig. 2 [29]. Biochar is produced by pyrolysis of biomass such as wheat straw or bamboo under limited oxygen conditions [42]. Its structure contains interconnected macropores (>50 nm) and mesopores (2–50 nm), which allow dissolved contaminants in soil solution to diffuse rapidly into the solid matrix. When MOFs such as MIL-100(Fe), UiO-66, or ZIF-8 are synthesized in situ within these pores, they introduce a large number of micropores (<2 nm). This creates a hierarchical pore system; larger pores enhance mass transport, while micropores provide high-energy adsorption sites. This structure improves

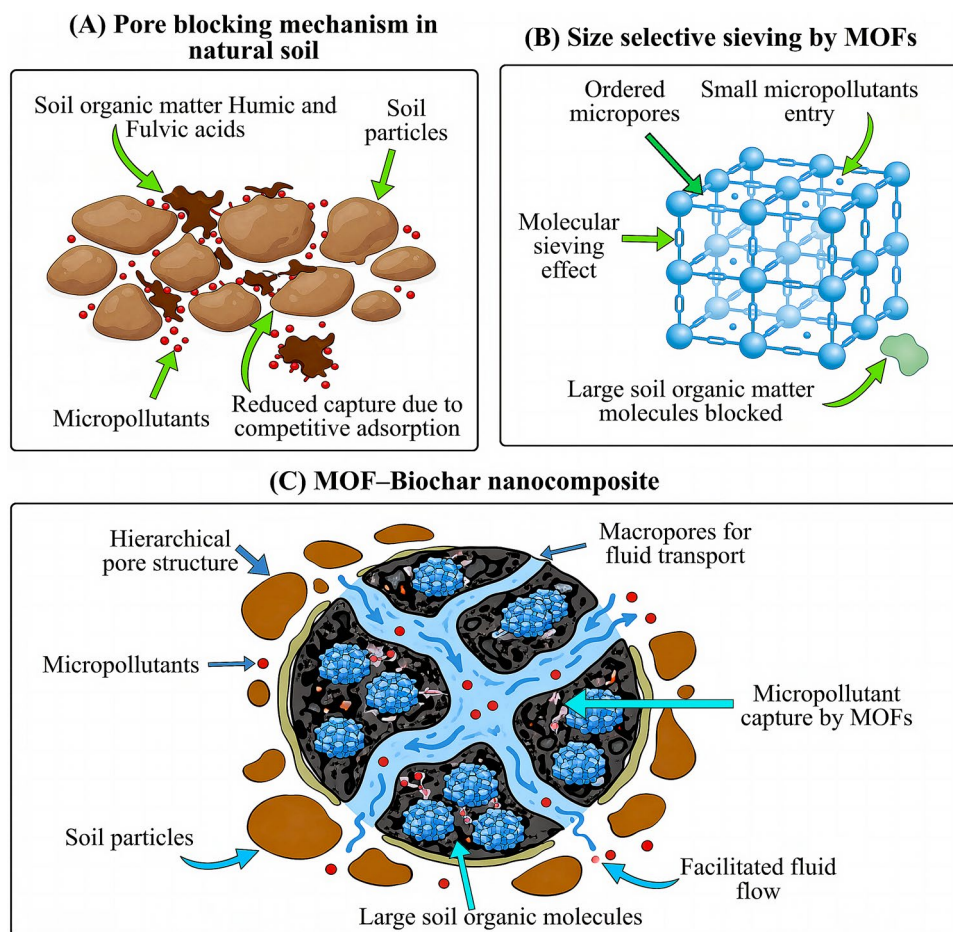


Fig. 2 Conceptual schematic representation of micropollutant capture in soil, MOFs, and MOF-biochar systems. **(A)** Competitive adsorption in natural soil reduces pollutant uptake. **(B)** MOFs enhance selective sieving through ordered microporosity. **(C)** MOF-biochar composites improve transport and overall capture efficiency

both diffusion efficiency and contaminant adsorption capacity [29]. The enhancement of adsorption performance may be realized through the use of MOFs in combination with biochar to form MOF-biochar nanocomposites. These materials form a hierarchical pore structure, in which biochar macropores are used to improve fluid transport, and MOF micropores are used to offer selective adsorption sites, with a pore volume of about $0.194 \text{ cm}^3 \text{ g}^{-1}$. The dual-pore system leads to an increase in the removal efficiency of complex soil contaminants compared with pure biochar, as illustrated in Fig. 2 [29]. Moreover, structurally stable structures like CALF-25 retain their pore sizes ($4.6 \text{ \AA} \times 3.9 \text{ \AA}$) at 90% relative humidity. This mechanical stability helps to avoid pore collapse and to maintain the same level of contaminant sequestration in realistic agricultural conditions [43].

The specific surface area of the WSB@MIL-100 composite is six times higher than that of raw biochar, and it is $419 \text{ m}^2 \text{ g}^{-1}$, and the total pore volume is $0.194 \text{ cm}^3 \cdot \text{g}^{-1}$ [29]. The nitrogen adsorption-desorption isotherm exhibits Type I behavior, which implies that the MOF phase is the dominant microporosity [44]. An H4-type hysteresis loop is also observed, which proves that the mesoporous structure of biochar is preserved. This equal distribution of pores minimizes the bottleneck effect that is usually realized in purely microporous materials, where low diffusion rates lower the adsorption efficiency [29].

Compared with similar materials, the improvement is significant. Rice straw biochar combined with MIL-100(Fe) has a surface area of $133.9 \text{ m}^2 \cdot \text{g}^{-1}$, and bamboo biochar combined with MIL-100(Al) reaches $44 \text{ m}^2 \cdot \text{g}^{-1}$. Although lower than the $557 \text{ m}^2 \cdot \text{g}^{-1}$ reported for ZIF-8/bamboo biochar pyrolysis hybrids, the $419 \text{ m}^2 \cdot \text{g}^{-1}$ obtained for WSB@MIL-100 provides an effective balance among porosity, structural stability, and scalability [8, 29].

The hybrid material contains a higher density of functional groups than either biochar or MOF alone. Biochar contributes oxygen-containing functional groups such as hydroxyl ($-\text{OH}$) and carboxyl ($-\text{COOH}$) groups. The MOF phase provides coordinated metal centers that act as specific adsorption and complexation sites. Synchrotron-based X-ray absorption fine structure and extended X-ray absorption fine structure analyses confirm that Cu^{2+} ions coordinate with four equatorial and two axial oxygen atoms, with bond distances ranging from 1.16 to 2.02 Å. This coordination structure shows the formation of stable surface complexes [29]. MIL-100(Fe)-modified biochar has a sorption capacity of about twice that of unmodified biochar, immobilizing 142 mmol kg^{-1} of Cu^{2+} and 156 mmol kg^{-1} of Pb^{2+} under soil conditions. The immobilization process involves the complexation of the surface and ion exchange, as indicated by measurable decreases in exchangeable Na^+ , Mg^{2+} , K^+ , and Ca^{2+} concentrations after treatment [29]. In general, the MOF-biochar composite is a structurally stable and chemically active soil remediation material. It combines high specific surface area, hierarchical porosity and increased metal-binding capacity into one engineered system, leading to increased contaminant immobilization efficiency.

2.3 Heavy metal immobilization through coordination

MOFs can reduce the concentration of heavy metals in soil because MOFs can be designed to include chemical groups with high affinity to toxic metal ions. Unlike traditional soil amendments, which mostly act on the surface, MOFs work both on the surface and inside the pores, which minimizes the amount of heavy metals available for plant uptake [29, 45]. Unsaturated metal sites and Lewis basic groups capture and form complexes with toxic metal like As, Cd, and Pb. The organic linkers of MOFs such as ZIF-8 and ZIF-67 provide pairs of electrons that strongly interact with Cd and make ions form stable complexes that do not migrate to the groundwater [46]. Iron-based MOFs, such as MIL-101(Fe), are highly efficient in the case of As, as Fe sites can substitute the ligands with As species, transforming mobile As into a stable mineral-like form in MOF pores and making it less toxic to crops [8]. Some MOFs are also applied alongside magnetic materials, where the contaminated particles can be easily collected after remediation and decrease the risk of secondary contamination [47].

The long-term effectiveness of MOFs is determined by their stability in agricultural soil under different redox conditions because of irrigation and microbial activity [29]. MOFs, such as CALF-25, contain phosphonate monoester linkers that create a hydrophobic environment, preventing the hydrolysis of coordination bonds and stabilizing the structure over a wide pH range, so that the metals trapped in them are not released into the soil during seasonal changes [43]. Even when MOFs are degraded slowly, they can form stable secondary minerals, such as goethite or hematite, through the oxalate carbonate route, which also entraps heavy metals and has long-term remediation effects

[26]. Supporting experimental data of MOF adsorption capacities for soil pollutants from conducted studies are further summarized in Table 1.

The comparative evaluation of the studies summarized in this review shows that the MOF incorporation into hybrid or composite systems has improved the adsorption performance. In several reports, MOF materials were found to have high adsorption capacities for pesticides and heavy metals. Although many pesticide-related values (e.g., loading capacities and controlled-release efficiencies) originate from agrochemical delivery studies, they demonstrate the strong host–guest interactions and tunable pore structures that also govern pollutant adsorption and sequestration in remediation applications.

In addition to surface area, the adsorption performance of MOFs can be influenced by a variety of other factors such as pore size, pore accessibility, surface functional groups, metal coordination, and structural stability. MOFs with appropriate pore structures and accessible active sites can promote the diffusion and adsorption of contaminants, and functional groups like amino, hydroxyl, and carboxyl groups can increase the interaction between the MOF and pollutants. The synthesis methods, such as linker modification and composite formation, may affect the adsorption selectivity, stability, and reusability. MOFs offer more structural tunability and multifunctionality than conventional adsorbents like biochar, activated carbon, zeolites, clay minerals, and engineered nanoparticles, but conventional materials are typically more cost-effective and scalable. Thus, the adsorption ability of MOFs is probably related to the structural and chemical characteristics of the MOFs, not to one specific property. The regeneration efficiency, adsorption–desorption performance and structural stability of MOFs in repeated use are crucial for their practical application. While many MOFs are able to achieve high adsorption efficiency over multiple regeneration cycles in the laboratory, the long-term environmental performance of these materials could be affected by changes in soil pH, redox potential, salinity, phosphate concentration, flooding, and microbial activity, both of which have the potential to impact framework stability and contaminant remobilization. Moreover, the environmental stability of MOFs is different, like UiO-66 and CALF-25 are more chemically stable than ZIF-8, HKUST-1 and MIL-101 in acidic or phosphate-rich environments. Thus, long-term leaching and field studies are needed to assess the environmental safety and applicability of the various MOF systems.

To facilitate future comparisons, we propose the Hybrid Performance Index (HPI) as a conceptual framework rather than a validated index. The HPI may integrate adsorption performance, structural stability, regeneration, and environmental compatibility into a unified assessment. However, because standardized datasets are currently unavailable, no universal calculation method can yet be established, and the HPI should be regarded as a hypothesis for future validation.

3 MOF-microbiome synergies for bioremediation

Biological restoration of polluted soil is frequently constrained by the low survival rates of specialized microbial inoculants in hostile, competitive, or polluted environments [51]. The combination of MOFs and beneficial microorganisms forms a synergistic platform in which the MOFs offer physical protection and a local nutrient reservoir, and the microbes execute specific metabolic degradation of pollutants [52, 53].

Table 1 MOF adsorption capacities for soil pollutants and ecotoxicological safety benchmarks

Interaction category	MOF	Target pollutant/analyte	Dominant adsorption/interaction mechanism	Adsorption capacity/value	Associated functional outcome	Reference
Pesticide sorption	UiO-66-NH ₂	Acetamiprid	Hydrogen bonding, electrostatic interaction	90.79%	UV-shielding efficiency	[19]
Pesticide sorption	ZIF-8	Lambda-cyhalothrin	Pore adsorption, hydrophobic interaction	87.71%	72 h sustained release	[35]
Pesticide sorption	MIL-101(Fe)	AZOX	Coordination interaction, pore adsorption	19.80%	Loading efficiency	[37]
Pesticide sorption	UiO-66	Diniconazole	Hydrogen bonding, π - π interaction	15.30%	92% fungal inhibition	[48]
Pesticide sorption	ZIF-8	Boscalid	Coordination interaction	25.10%	Coordination efficiency	[46]
Pesticide sorption	MIL-101(Fe)	Emamectin benzoate	Hydrogen bonding, pore adsorption	14.20%	Improved foliar adhesion	[49]
Metal sequestration	WSB@ MIL-100	Cu ²⁺	Surface complexation, coordination	142.00 mmol kg ⁻¹	96.5% removal rate	[29]
Metal sequestration	WSB@ MIL-100	Pb ²⁺	Surface complexation, coordination	156.00 mmol kg ⁻¹	Hierarchical pore capture	[29]
Metal sequestration	Fe-MOF (OPA)	AsO ₄ ³⁻	Ligand exchange, coordination	50.00 mg kg ⁻¹	Below FAO safety limit	[7]
Metal sequestration	Fe ₃ O ₄ @ UiO-66	Heavy metals	Coordination, electrostatic interaction	96.00%	Magnetic recovery rate	[47]
Metal sequestration	ZIF-67	Cd ²⁺	Coordination interaction	120.50 mg g ⁻¹	Pore-surface coordination	[46]
Ecotoxicity	ZIF-8	<i>Eisenia fetida</i>	Not reported	>1000 mg kg ⁻¹	No mortality observed	[8]
Ecotoxicity	MIL-100(Fe)	Soil microbiome	Not reported	15.00%	Evenness index increase	[29]
Ecotoxicity	ZIF-8	Pollinators (bees)	Not reported	LD ₅₀ >100 μ g bee ⁻¹	Encapsulation safety	[50]
Ecotoxicity	Fe-MOF (OPA)	Rice plant	Not reported	18.00%	Chlorophyll increase	[16]
Ecotoxicity	Mg-MOF	Maize plant	Not reported	64.90%	Photo-synthetic stability	[27]
Soil physics	MOF-Biochar composite	Water holding capacity	Pore structure enhancement	2.95%	Increase per % SOC	[29]

Table 1 (continued)

Interaction category	MOF	Target pollutant/analyte	Dominant adsorption/interaction mechanism	Adsorption capacity/value	Associated functional outcome	Reference
Soil physics	MOF nano-carrier	Soil porosity	Not reported	<30.00%	Reduced clogging	[16]
Leaching risk	ZIF-8	Zn ²⁺	Controlled ion release	<0.50 mg L ⁻¹	Below WHO drinking limit	[8]
Leaching risk	UiO-66	Particle mobility	Not reported	<5.00 cm	14-day vertical transport	[49]
Surface chemistry	PDA@ZIF-8	Contact angle	Surface functionalization	45.20 degree	Enhanced leaf wetting	[50]
Surface chemistry	CALF-25	Surface area	Permanent porous framework	2100 m ² g ⁻¹	Permanent porosity	[43]

3.1 The exoskeleton paradigm for microbial protection

In situ biomimetic mineralization is the basis of the idea of a microbial exoskeleton, where a thin layer of MOF is deposited on the surface of single bacterial cells. The process forms a conformal and semi-permeable shell that controls the exchange of molecules between the cell and the external environment [54, 55]. Bacterial strains that can degrade pollutants, such as those that metabolize p-nitrophenol (PNP), can be encapsulated in a ZIF-8 shell and remain metabolically active. The oligotrophic conditions of 7 days preserve the viability of encapsulated cells by 79%, but the viability of unprotected cells usually decreases rapidly during the same period [21]. The microporous ZIF-8 structure is a molecular sieve. It prevents the diffusion of large toxic macromolecules and lytic enzymes but does not prevent the diffusion of smaller pesticide molecules into the cytoplasm, where enzymatic biodegradation occurs, as shown in Fig. 3.

There are evident performance variations in the soil systems. ZIF-8-encapsulated cells in unsterilized black soil fully degraded 100 mg kg⁻¹ PNP in 5 days. The same encapsulated system in acidic red clay (pH 5.2) resulted in 100 mg kg⁻¹ being fully degraded in 12 days. Conversely, native uncoated cells deteriorated less than 20 mg kg⁻¹ in the same acidic soil matrix in 14 days [21]. The mineral coating also increases the mechanical stability and ecological performance. *Novosphingobium* cells coated with ZIF-8 have a higher capacity for adhesion and colonization of roots. Root-associated populations of colony-forming unit (CFU) assays are 1.54×10^7 to 2.08×10^7 CFU g⁻¹ in comparison with 0.17×10^7 to 0.30×10^7 CFU g⁻¹ of bare MOF materials. The improved colonization is linked to measurable changes in saline alkali soils. The pH of the soil reduces by 0.6, from 8.5 to 7.9, and the salt content reduces by 0.11 g kg⁻¹, from 4.3 to 3.1 g kg⁻¹ [56]. The root malondialdehyde levels are reduced by 42.00%, and proline levels are reduced by 39.13%, which means that the oxidative stress and osmotic stress in the rhizosphere are reduced [56]. MOF nucleation takes place by coordination of metal ions with functional groups on the bacterial cell surfaces, creating a porous protective shell. This semi-permeable exoskeleton allows nutrient and metabolite diffusion and at the same time allows microbial metabolic activity, which slowly breaks down by hydrolysis, enzymatic activity and pH changes in the rhizosphere. There is currently evidence indicates of compatibility with a variety of microbial species, but species-specific performance needs to

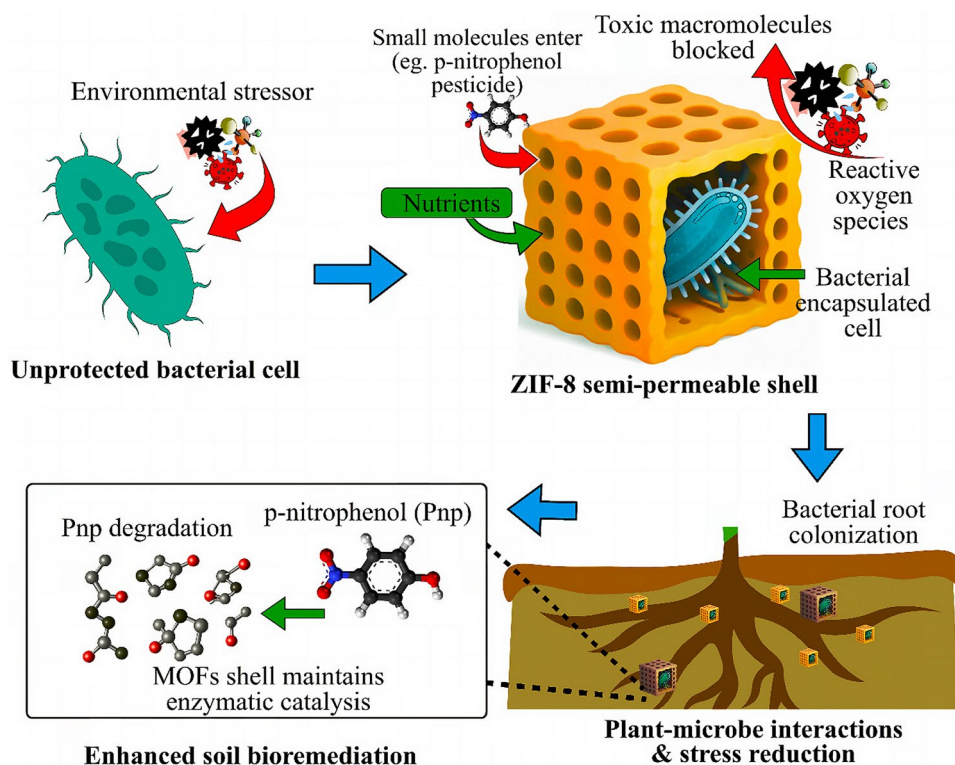


Fig. 3 Conceptual illustration of MOF-mediated encapsulation of bacteria shows selective permeability, stress protection, sustained catalytic activity, and enhanced plant-microbe interactions for improved soil remediation

be investigated. Collectively, these results indicate that mineral-encapsulated bacterial cells are metabolically active and acquire enhanced resistance to environmental stressors, which allows them to maintain more stable biodegradation activity and enhanced plant-microbe interactions under field conditions. Table 2 further summarizes the quantitative data on how MOFs can enhance the protective performance of microbes against a range of agrochemicals.

The studies summarized in Table 2 indicate that loading efficiency and microbial viability can vary considerably among different MOF systems. These differences may reflect variations in MOF composition, cargo properties, and encapsulation approaches, which can influence controlled release, microbial protection, and overall delivery performance. Collectively, these findings suggest that optimizing both MOF design and encapsulation strategy may be important for improving agricultural applications.

3.2 MOF-assisted bio-augmentation in degraded soils

To achieve effective bio-augmentation of degraded soils, the introduced microorganisms must be able to withstand environmental stress, migrate to contaminated microsites, and be able to stabilize their population in the rhizosphere [59]. MOF-based delivery systems offer a structurally defined system of microbial transportation. These materials protect cells when applied to the soil and enhance spatial localization at the soil-root interface [60]. Zeolitic imidazolate frameworks have been utilized as protective carriers for specialized degraders. Encapsulation alleviates osmotic and oxidative stress in oligotrophic soil [21]. This finding shows that the microporous matrix balances environmental changes without altering metabolic activity. Affinity towards root exudate matrices is

Table 2 Quantitative performance of MOF-based platforms for pesticide delivery and microbial protection

Category	MOF type	Cargo/microbe	Loading/encapsulation (%)	Concentration/dosage	Reported functional outcome	Reference
Fungicide	ZIF-8	Boscalid	25.10%	50 mg L ⁻¹	78.5% inhibition (<i>B. cinerea</i>)	[46]
Fungicide	Fe-MOF (MIL-101)	AZOX	19.80%	100 mg L ⁻¹	pH-responsive release (pH 5.0)	[37]
Fungicide	Zr-MOF (UiO-66)	Diniconazole	15.30%	200 µg mL ⁻¹	92% fungal inhibition	[48]
Fungicide	Fe-MOF (CMC-coated)	Difenoconazole	22.40%	10 mg kg ⁻¹	76.16% cellulase-triggered release	[21]
Fungicide	ZIF-8	Tebuconazole	18.90%	5.0 mg L ⁻¹	Reduced non-target toxicity	[50]
Insecticide	ZIF-8	Chlorpyrifos	12.50%	0.5 mg cm ⁻²	100% mortality (<i>P. xylostella</i>)	[8]
Insecticide	UiO-66-NH ₂	Acetamiprid	90.79%	1.0 g L ⁻¹	UV-shielding protection	[19]
Insecticide	MIL-101(Fe)	Emamectin benzoate	14.20%	25 mg L ⁻¹	Enhanced foliar adhesion	[49]
Insecticide	ZIF-8	Lambda-cyhalothrin	87.71%	0.2% (w/w)	Reduced soil leaching	[35]
Insecticide	UiO-66	Thiamethoxam	21.00%	150 mg L ⁻¹	85% systemic uptake (rice)	[17]
Herbicide	MIL-100(Fe)	Glyphosate	28.50%	2.0 kg ha ⁻¹	Minimized phosphate competition	[26]
Herbicide	UiO-66	Atrazine	11.80%	1.5 mg L ⁻¹	95% adsorption in groundwater	[29]
Bioprotection	ZIF-8	<i>Novosphingobium</i>	79.00% (viability)	10 ⁸ CFU mL ⁻¹	30% increase in salt tolerance	[56]
Bioprotection	ZIF-8	<i>Bacillus subtilis</i>	82.00% (viability)	10 ⁷ CFU mL ⁻¹	Survival at 50 °C thermal stress	[19]
Bioprotection	ZIF-8	<i>Pseudomonas</i> spp.	75.00% (viability)	10 ⁹ CFU mL ⁻¹	UV-B resistance (254 nm)	[8]
Bioprotection	Fe-MOF	<i>Azotobacter</i>	68.00% (viability)	2.5 g kg ⁻¹ soil	Enhanced N ₂ fixation rate	[7]
Physiology	Mg-MOF	Magnesium ions	100.00% (node-based)	50 mg kg ⁻¹	64.9% increase in Mg uptake	[57]
Physiology	ZIF-8	Zinc ions	100.00% (node-based)	10 mg L ⁻¹	62.96% root weight increase	[58]
Remediation	MOF–Biochar composite	Cu ²⁺	142 mmol kg ⁻¹ (adsorption)	100 mg L ⁻¹	96.5% heavy metal removal	[29]
Remediation	MOF–Biochar composite	Pb ²⁺	156 mmol kg ⁻¹ (adsorption)	100 mg L ⁻¹	Hierarchical pore sequestration	[29]

enhanced by modification with synthesized polysaccharide-binding proteins (SPBP). Zn signal intensities of 26,695–36,694, in contrast to 2,554–5,002 in non-modified controls, in wheat co-culture systems demonstrate improved localization of SPBP-functionalized ZIF-8 assemblies (ZSBet) to the root surface [56].

Unlike bulk inoculation, which can frequently be restricted by rapid dilution, predation, and dispersal, MOF-microbe co-assemblies can facilitate gradual release and long-term colonization [61]. CFU assays demonstrate that the root-associated bacterial densities of 1.54×10^7 to 2.08×10^7 CFU g⁻¹ occur after MOF-assisted delivery. Traditional inoculants are limited to 0.17×10^7 to 0.30×10^7 CFU g⁻¹ under comparable circumstances [56]. The higher the colonization density, the higher the rate of contaminant degradation. When 100 mg kg⁻¹ PNP is added to unsterilized black soil, full degradation takes place in 5 days. When the red clay soils are acidic, it takes 12 days to remove all the 100 mg kg⁻¹. In the same conditions, native cells degrade less than 20 mg kg⁻¹ within the same time which proves the kinetic superiority of structured delivery [21].

In addition to transport and localization, MOF scaffolds facilitate the development of biofilms that are mechanically stable. ZSBet + *Novosphingobium* co-assemblies in saline-alkali stress rhizospheres of wheat lower soil pH (8.5) to 7.9 (−0.63 units) and salt content (4.3 to 3.1 g kg⁻¹; -1.2 g kg⁻¹) [56]. The changes are also accompanied by the increase in soil organic matter and the availability of nitrogen and phosphorus. The grain number is raised by 145.05%, which is a result of remediation and growth-promotion of the plant. MOF-based composites can also reduce metal toxicity that inhibits microbial growth during bio-augmentation. Adsorption experiments have demonstrated sorption capacities of approximately 142 mmol kg⁻¹ for Cu²⁺ and 156 mmol kg⁻¹ for Pb²⁺ in contaminated soils [29]. Taken together, MOF–microbe assemblies integrate physical protection, spatial targeting, controlled release, and nutrient co-delivery into a single engineered system. This approach has the potential to enhance bioaugmentation into a spatially regulated and mechanistically coordinated process, linking microbial persistence, contaminant degradation, and plant productivity within the same microenvironment.

3.3 Metabolic support of pollutant-degrading microorganisms by MOFs

MOFs may also play a direct role in the metabolism of soil microorganisms other than protection. These materials can be used as temporary nutrient stores when they are synthesized using biologically associated ligands and metal nodes [53]. They provide the necessary factors that sustain microbial activity in the process of bioremediation, especially in oligotrophic soils with low carbon and mineral levels [55].

Iron-based MOFs with phosphorus- and nitrogen-containing ligands, including those of empirical composition C₂H₁₅Fe₂N₂O₁₈P₃, contain about 6.03% N, 14.48% P, and 14.69% Fe [26]. These elemental fractions enable the framework to serve as a multi-nutrient source. Nutrients are released gradually as the crystalline structure is gradually biodegraded, instead of being released in one pulse. Fe-MOFs exhibit 82% structural degradation in soil systems in 98 days, which is significantly faster than in aqueous systems, suggesting that mineral dissolution in soil systems is mediated by microbes [26]. Nutrient release follows a comparable temporal pattern. After 98 days, mineral N, P, and Fe have cumulative release of 85, 75, and 34%, respectively [26]. This intermittent release is in line with the growth needs of microbes and the dynamics of plant nutrient uptake. MOFs thus combine nutrient provision and contaminant elimination through

controlled mineral decomposition. This synchronized mechanism establishes a chemically integrated system where the bioremediation and metabolic support take place simultaneously [21, 56]. The compatibility of MOFs and microorganisms might rely on the microbial properties, MOF composition, and environmental conditions. Strains that form more biofilm or are more tolerant to metal ions could show better interactions with MOFs. However, their performance under field conditions may be influenced by environmental variability and competition with native soil microorganisms. Therefore, further field studies are needed to evaluate the stability and effectiveness of MOF–microbiome systems in agricultural soils. In addition to their physicochemical sorption properties, MOFs are now being shown to have a biological effect on soil, indicating the formation of an integrated network of MOF–microbiome–plant interactions. The use of MOF formulations, especially those that deliver agrochemicals or micronutrients in a controlled manner, has been reported to enhance microbial activity and plant growth in several studies reviewed here. Based on the available evidence, we propose the Rhizosphere Engineering Triangle as a conceptual framework describing the potential interactions among pollutant immobilization, microbial activity, and plant nutrient uptake. Rather than representing a validated model, it is intended to guide future hypothesis development and experimental investigation.

4 Biologically driven dissolution of MOFs

The biologically mediated disassembly processes control the transformation of MOFs into functional remedial agents into integrated soil components [62]. In contrast to synthetic polymers that can remain in the form of microplastics, the structural units of specially designed MOFs are to be subjected to microbial mineralization and enzymatic degradation to enter natural biogeochemical cycles [63, 64].

4.1 Microbial mineralization of organic linkers

MOFs structure is determined by organic linkers and constitutes the primary interface between the soil microbial communities and the MOFs. Such carbon-rich ligands are chemically active in the soil [18]. Once they have been applied, they are subjected to extracellular enzymes and microbial metabolism. Their change regulates the stability of the framework as well as the cycling of nutrients [65]. Advanced design of MOF-based agrochemicals includes the integration of certain chemical bonds that react to biological catalysts or other environmental alterations, such as Schiff base linkages, natural polyamine coordination, biodegradable polymer conjugation, and phosphate-amine systems. As an example, acid-sensitive Schiff base (imine) bonds in structures like ZIF-90 can be introduced to facilitate multimodal antimicrobial action; in the presence of acidic conditions caused by the infection of *Magnaporthe oryzae*, the bonds break down, releasing kasugamycin (KSM) and Zn^{2+} ion as illustrated in Figure. 4 [48].

In iron-based MOFs with oxalate-based linkers, degradation proceeds primarily via the oxalate–carbonate pathway. Half the coordination network of metal–ligand is hydrolyzed, and oxalate is liberated into the soil solution. Oxalate is then used as a source of carbon and energy by oxalotrophic bacteria [26]. This biological consumption changes dissolution equilibria and enhances structural disintegration. In aqueous systems, the release of nutrients does not exceed 10%. Conversely, the soil systems encourage rapid mineralization, and up to 82% of the mass can be lost in 98 days [26]. The difference

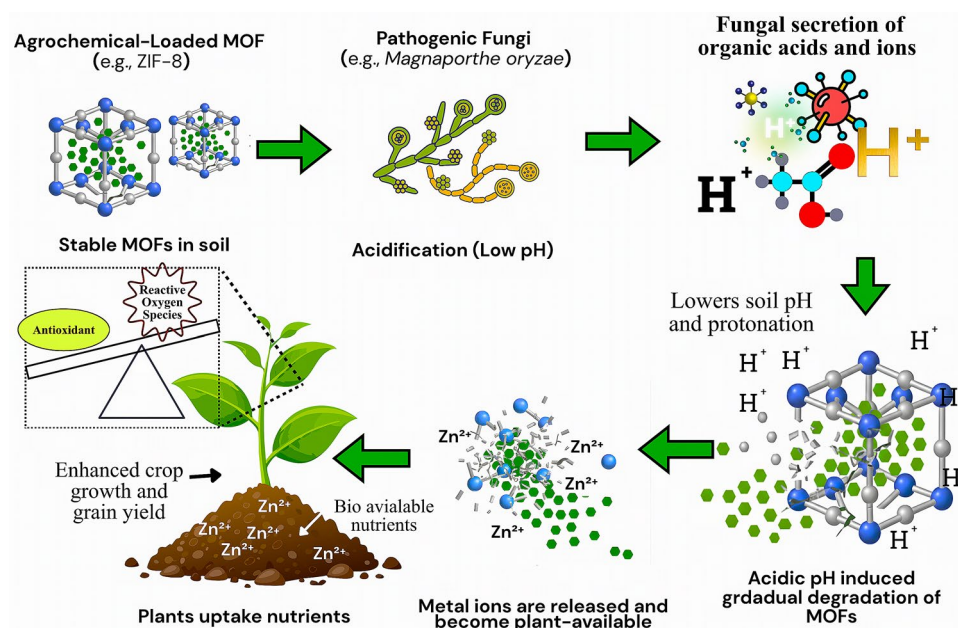


Fig. 4 Conceptual illustration of fungi-triggered MOF degradation for controlled agrochemical release. Fungal secretions cause lower soil pH which disintegrate MOFs and nutrients are available for plants uptake, while minimizing agrochemical loss and environmental impact

indicates the active catalytic action of the soil microbiome in the degradation of MOF as illustrated in Fig. 5. Similar soil-enhanced release patterns have been observed for other coordinated nutrients. During the same incubation period, cumulative phosphorus release reached 75 and 68%, while iron release reached approximately 34–35% in soil environments [26]. These values indicate that MOF decomposition not only disassembles the framework but also gradually transfers coordinated nutrients into plant-available soil pools. The biodegradation rate of MOFs varies with their metal nodes, organic linkers, and soil conditions. Some of the biodegradable MOFs degrade quickly, while others are more stable and can remain in place for a longer period of time, keeping the contaminants immobilized throughout the growing season. Released metals can also be bound to soil minerals or organic matter, which can result in continued contaminant retention even after partial framework degradation.

Ligand turnover is greatly affected by temperature. In a wheat growing season, the rate of degradation rose to 13.8% at 19.8 °C and 4.2% at 4.7 °C monthly, which indicated thermally stimulated microbial and chemical activity [16]. After more than six months, soils to which these MOFs were added had a higher concentration of mineral N, available P, and Fe than the untreated controls and urea-fertilized soils [16]. Linker mineralization thus helps in long-term nutrient supply as the structure breaks down. The oxalate-carbonate pathway also influences the soil pH. Localized buffering effects can be generated by microbial oxidation of oxalate to carbonate, and protons can be produced by oxidation of Fe^{2+} to Fe^{3+} . Experimental evidence indicates that the observable pH shifts are primarily observed during the first two months of application, and then stabilization is observed when redox balance and carbonate equilibria are restored [7, 16]. These redox and carbon reactions are coupled and combine MOF degradation into more extensive soil mineral and biogeochemical cycles, as shown in Fig. 5.

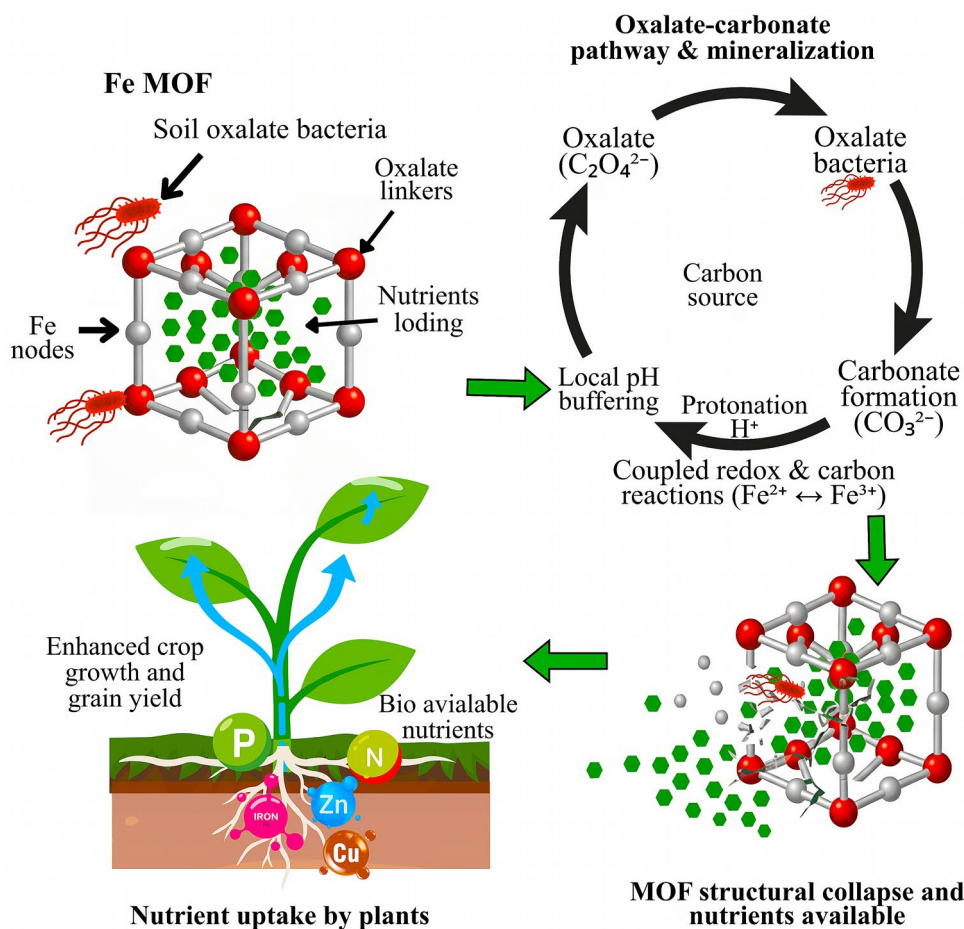


Fig. 5 Conceptual visualization of Fe-MOF microbe interactions in soil for nutrient release and plant growth. The diagram shows the interaction between oxalate-producing soil bacteria with Fe-based MOFs to transform minerals, regulate framework disintegration, and enhance nutrient accessibility to plants

MOFs with amine- or carboxylate-functionalized linkers are enzyme-degradable as well. A carboxymethyl cellulose (CMC) coating allows enzyme-responsive behavior in NH_2 -MIL-101(Fe). A cumulative cargo release of 76.16% is achieved after 120h under cellulase-active conditions at pH 7.0, whereas the cumulative cargo release is about 55% in the absence of enzyme activity [21]. Structural degradation of amino-functionalized materials, including NH_2 FeMIL101 is further enhanced by acidic conditions. Encapsulated diniconazole is released at a rate of up to 97% in 100h at pH 3.1, which is in line with partial degradation of polydopamine or ligand-derived surface layers [50].

Organic components are added to soil organic matter pools by progressive ligand mineralization and coordinated metal ions are partitioned into exchangeable or mineral-associated fractions. This combined nutrient delivery has been linked to grain yields that are nearly twice that of phosphorus-deficient controls in iron-based OPA-MOF systems [7]. Hence, MOF ligand degradation is not only structural degradation but also a controlled part of soil carbon processes, nutrient supply, and crop, as shown in Fig. 5. The addition of organic components to pools of soil organic matter promotes progressive mineralization of ligands and the partitioning of coordinated metal ions into exchangeable or mineral-bound fractions. These joint nutrient delivery in Fe-based OPA-MOF systems have been associated with a twofold increase in grain yield over

phosphorus-deficient controls [7]. Thus, the degradation of MOF ligands is not structural degradation but also a regulated contribution to soil carbon cycles, nutrient cycling and crop yields. Microbial degradation can also be affected by porosity and surface properties of MOF-derived composites. As an example, MOF biochar nanocomposites in soil remediation systems have specific surface areas of about $419\text{ m}^2\text{ g}^{-1}$, and pore volumes of about $0.194\text{ cm}^3\text{ g}^{-1}$, which offer a large surface area of contact between microbial cells and framework structures [29]. These highly porous architectures enhance the exposure of microbial enzymes to organic linkers and hence promote biologically mediated dissolution.

MOFs have a high potential for soil remediation but their environmental fate and long-term safety are important considerations for field deployment. There is currently no evidence that they are ecotoxic or biodegradable, which is dependent on the composition of the metal nodes, the particle size, and degradation behavior. The long-term environmental compatibility of Zr- and Cu-based MOFs should be assessed. In contrast, Fe-, Mg-, and Ca-based MOFs are generally considered more suitable for sustainable agriculture applications, since their degradation products can be beneficial for natural nutrient cycles, but have lower ecotoxicological risks. In this regard, the design of MOFs in the future should focus on the use of environmentally friendly metal nodes, biodegradable linkers, and controllable degradation to ensure high remediation efficiency and long-term ecological safety. To support future environmental risk assessment, we propose the Environmental Safety Threshold (EST) as a conceptual framework rather than a standardized safety index. The conceptually derived EST could be used to represent concentration ranges where MOFs do not cause measurable adverse ecological effects under defined conditions. More laboratory and field research is needed, however, before quantitative thresholds can be determined.

4.2 Environmental factors influencing the degradation kinetics

The degradation of MOFs in agricultural soil is not a fixed process but is very sensitive to the abiotic factors of the soil matrix [28]. Soil moisture and soil temperature are the main external factors that control the hydrolytic stability of coordination bonds and the metabolic activity of the microbial communities that mediate linker mineralization as illustrated in Fig. 6. The stability of MOFs is a major factor that is controlled by soil moisture [66]. Water is used as a reactant in the hydrolytic cleavage of metal-ligand bonds, and as a medium of transport of dissolved metal ions and organic ligands [67]. Field moisture leads to an extreme structural transformation in iron-based materials such as Fe-MOF (OPA). A wheat growing season has been reported to lose 50.9% of its mass, a gradual degradation of the framework in the natural pedoclimatic conditions [16]. Water molecules move into the pore structure and organize the metal nodes destabilizing coordination bonds and initiating hydrolysis. This process facilitates the continuous release of nutrients in moist conditions, which are maintained. The levels of minerals like N, available P, and Fe in MOF-amended soils are significantly greater than the urea-treated controls after six months, which demonstrates a slow release of nutrients and long-term retention [16].

The soil water content also changes, enhancing degradation. The drying and wetting cycles lead to alternating chemical and mechanical stress [68]. Drying leads to an increase in the concentration of solutes and an increase in the ionic strength of the soil

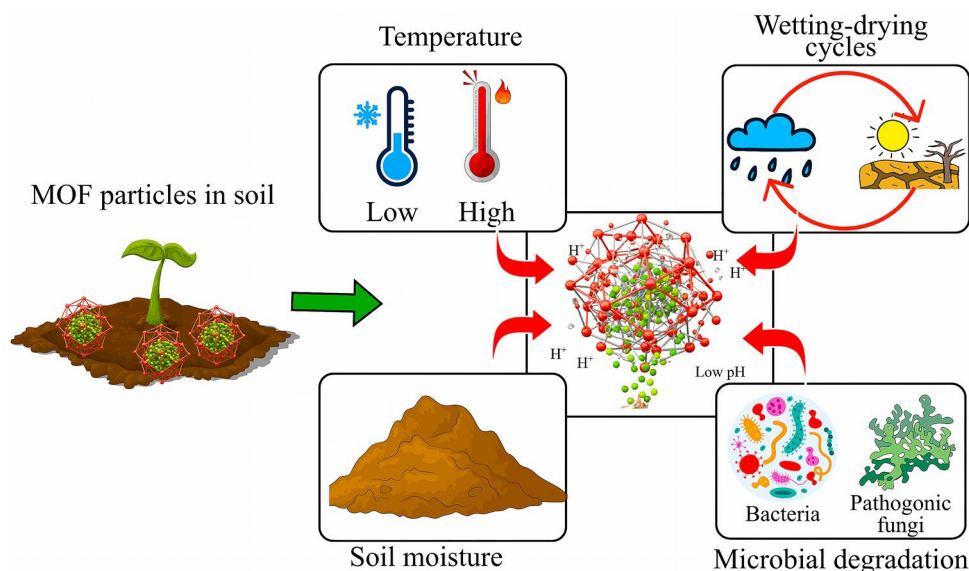


Fig. 6 Environmental factors governing MOF degradation in soil systems

pores, whereas rapid rewetting results in the formation of osmotic gradients and volumetric expansion [69]. These repetitive alterations promote the formation of microfractures and dissociation of ligands. Some MOFs have been chemically modified to increase the hydrolytic stability. As an example, the functional groups of ethyl esters in CALF-25 sterically hinder the water access to the hydrolysis-sensitive coordination sites. The structural integrity of this design is 90% at 353 K and a relative humidity of 90% [43]. The temperature is in contact with the moisture to regulate the persistence of MOF. According to the field data, the correlation between the soil temperature and the monthly rate of Fe-MOF degradation is nearly linear ($R^2 = 0.99$). The degradation rate is approximately 4.2% per month at 4.7 °C. At 19.8 °C, the rate increases to 13.8% per month, which is more than three times higher than the rate observed at 4.7 °C. This temperature dependence suggests enhanced hydrolysis and increased microbial metabolic activity under warmer soil conditions [16].

MOF-based delivery systems can also be designed to be environmentally responsive. A thermoresponsive poly (N-isopropylacrylamide) shell regulates the release of pesticides in IDC@UiO-66-(COOH)₂-PNIPAm. The polymer has a lower critical solution temperature (LCST) of about 32 °C. At temperatures lower than this, diffusion of encapsulated indoxacarb (IDC) is inhibited by the hydrated polymer network. Polymer chain collapse above the LCST, which enhances pore permeability. Cumulative release of IDC is 79.0% at 40 °C and 26.58% at 15 °C [70]. This temperature-dependent permeability coincides with the release of pesticides with the times of increased temperature, which is usually accompanied by increased pest activity and crop metabolic demand. In general, the transformation of MOF is regulated by soil moisture and temperature in a complex of hydrolytic, mechanical, and biological processes, as illustrated in Fig. 6. MOF degradation is a climate-sensitive process in soil environments involving the interaction of material chemistry and dynamic physical and biological conditions of the rhizosphere, and not an abiotic decay process.

Based on the overall evidence, MOF transformation in soil should be understood in the context of a model of degradation that is responsive to the rhizosphere environment,

the rhizosphere-responsive degradation model (RRDM), instead of in the context of isolated abiotic hydrolysis. The degradation kinetics in this model are controlled by the dynamic interaction of climatic factors (temperature and moisture), soil physicochemical factors (ionic strength and redox changes), and biological factors (root exudation and microbial metabolism). All these factors work together and in a cyclic manner to create spatiotemporal “degradation hotspots” in the rhizosphere where hydrolysis, ligand mineralization and nutrient release are enhanced. The idea of MOF degradation as a climate- and biology-synchronized process offers a predictive approach to designing environmentally adaptive MOFs that can be tuned to the crop growth stages and seasonal soil dynamics. Such a model can serve as a basis for future field-scale prediction, life-cycle assessment, and design of materials that are resilient to climate change.

4.3 Predictive modelling of long-term persistence

To determine the long-term environmental compatibility of MOFs in agricultural soils, the residence time of these materials has to be estimated to avoid the unwanted accumulation of nanomaterials [28]. The kinetics of degradation is highly reliant on the soil texture and the SOC content [66]. High macroporosity and high hydraulic conductivity enhance water flow and oxygen diffusion in sandy or light-textured soils with low SOC ($<15.0 \text{ g kg}^{-1}$). Such circumstances favor hydrolytic fracture of metal–ligand bonds and microbial colonization of the exposed framework surfaces, leading to accelerated structural degradation compared to fine-textured soils [16].

Semi-empirical kinetic models are typically used to describe degradation and nutrient release. In the case of Fe-based MOFs, the cumulative release of N, P, and Fe has been modeled using the Higuchi model and the Korsmeyer–Peppas equation. The values of diffusion exponents (n) of 0.43 to 0.85 suggest non-Fickian, anomalous transport [26]. These values imply that the release of nutrients is regulated by a combination of processes, such as diffusion through the pores, erosion of the matrix by microorganisms, and degradation of ligands by microorganisms, and not by diffusion alone. Cargo release half-life and structural half-life should be differentiated. For example zirconium-based UiO-66-NH_2 releases the herbicide MCPA to 0.037 mg mL^{-1} in distilled water, and the crystalline structure is mostly preserved under the same conditions [17]. Conversely, ZIF-8 systems tend to release encapsulated compounds rapidly at relatively low acidic pH (3–5), and the release of a large cargo occurs within 12 h. Nevertheless, the Zn–imidazolate coordination network is more stable in neutral soils, in which hydrolysis is slower [71]. These variations indicate that active compound release kinetics and matrix persistence cannot be considered together in environmental risk assessment.

Structural and spectroscopic verification is necessary to achieve long-term transformation. Fe-based MOFs show progressive mineralization by reduction and eventual loss of typical powder X-ray diffraction (PXRD) reflections, and weakening of Fourier transform infrared (FTIR) bands of coordinated carboxylate groups, especially asymmetric and symmetric vibrations at 1650 and 1430 cm^{-1} [26]. During iron-based OPA-MOF soil amendments, the nitrate-N levels grow slowly and stabilize at about 200 mg kg^{-1} after 17 weeks, which is in line with the addition of framework-derived nitrogen into soil nutrient pools [7, 16].

The physicochemical characteristics of MOF-derived composites can influence long-term persistence and degradation behavior in soil systems. For example, biochar–MOF

composite materials have reported surface areas ranging from $41 \text{ m}^2 \text{ g}^{-1}$ for CBZ@UiO-66@ZnO/biochar, $219 \text{ m}^2 \text{ g}^{-1}$ for ZIF-67/biochar pyrolysis composites, $146 \text{ m}^2 \text{ g}^{-1}$ for BC/Mg-MOF-74 materials, and approximately $96 \text{ m}^2 \text{ g}^{-1}$ for MIL-53(Fe)/biochar systems [72, 73]. These structural differences directly affect diffusion pathways, microbial accessibility, and water penetration, which are critical parameters controlling long-term degradation rates and persistence of MOF structures in soil environments. To contextualize these outcomes, Table 3 summarizes experimental data on the environmental fate and persistence of MOFs in diverse soil and biological systems. Although laboratory results have been promising, the translation of MOF technologies to field-scale applications is still limited. Most studies are carried out in controlled conditions with simplified soil matrices that do not necessarily reflect the complexity of natural agricultural systems. MOF performance can be affected by factors like soil heterogeneity, competing ions, organic matter and climatic variability. While microbial degradation can facilitate the release of MOFs as bioavailable nutrients, the environmental fate of the released metal ions and organic linkers is not well understood. The accumulation of repeated applications can be affected by the MOF composition, application rate and soil properties. Furthermore, the degradation products may have ecotoxicological effects on soil microorganisms and non-target organisms that needs to be investigated under long-term field conditions.

5 Translocation and multi-trophic transfer

When MOFs break up in the soil matrix, the metal nodes and organic linkers are released into the soil solution and are accessible to biological uptake [74]. The mechanisms of these degradation byproducts within the plant and into the wider food web are crucial to assessing the effectiveness of biofortification as well as the safety of the agricultural product [75].

5.1 Plant mediated uptake of degradation byproducts

Plants constitute the primary biological route of the elements of MOFs to be moved out of soil into higher trophic levels. The uptake of elements is dependent on the chemical speciation and mobility of ions that are released during framework degradation [76]. Structural disassembly can create a long-term and bioavailable pool of nutrients against rapid release of highly soluble mineral salts. Following the breakdown of structures like Fe-MOF (OPA) or ZIF-8, coordinated metal ions, e.g., Fe^{3+} , Zn^{2+} , and Mg^{2+} , are discharged into the soil solution and into the known rhizosphere transport systems [27, 58].

The physiological activity of plants proves that these materials are sources of nutrients. In faba bean, fresh shoot biomass and dry shoot biomass are enhanced by 37.79% and 40.42% respectively when Cu-BTC is applied compared to the untreated controls. Root development is also improved and the fresh root weight and dry root weight increase by 62.96% and 75.0% respectively. The stem diameter grows by 31.5% and the leaf surface area grows by 46.22%, which means that the vascular development and photosynthetic capacity improve [58]. These growth effects are linked to a 1.91-fold growth in carotenoid concentration and a 2.07-fold growth in chlorophyll b content, which indicates stimulation of pigment biosynthesis and light-harvesting efficiency. As well as the GR-MOF-27 formulation enhances Mg uptake by 64.9% and P uptake by 57.4% compared to magnesium sulfate (MgSO_4) (Morales-Cámara, 2025). ZIF-8-based formulations with

Table 3 Environmental kinetics, degradation half-lives, and bioaccumulation of MOFs byproducts

Framework environment	MOF system	Analyzed parameter	Numerical value/percentage	Time interval/units	Observed effect	Reference
Paddy soil	Fe-MOF (OPA)	Degradation rate	82.0%	98 days	Conversion to CaCO ₃	[26]
Maize field	MIL-101(Fe)	Cumulative release	76.16%	144 h	Cellulase-triggered flux	[21]
Rhizosphere	ZIF-8	Zinc bioavailability	62.96%	30 days	Increased root biomass	[58]
Laboratory	GR-MOF-27	Magnesium uptake	64.9%	21 days	Enhanced photosynthesis	[27]
Aquatic system	UIO-66-NH ₂	Sustained release	92.0%	1152 h	pH 5.0 responsiveness	[49]
Foliar surface	IDC@UIO-66	Cargo retention	85.0%	72 h	Resistance to rain wash	[70]
Wheat soil	ZIF-8	Microbial viability	79.0%	15 days	Saline-alkali resilience	[56]
Sandy soil	Fe-MOF	Half-life (t _{1/2})	45 days	25 °C	Rapid mineral dissolution	[16].
Clay soil	Fe-MOF	Half-life (t _{1/2})	60 days	20 °C	Adsorption passivation	[16].
Rice tissue	MIL-100(Fe)	Iron translocation	35.0%	Post-harvest	Root-to-grain mobility	[26]
Faba bean	Cu-BTC	Copper uptake	12.4 mg kg ⁻¹	Dry weight basis	Within safety thresholds	[58]
Saline soil	ZIF-8	Na ⁺ exclusion	40.0%	20 days	Reduced ion toxicity	[56]
Rhizosphere	Fe-MOF	P-solubilization	57.4%	45 days	Increased nutrient availability	[27]
Soil solution	CALF-25	Water stability	90% RH	7 days	Pore structure retention	[43]
Maize root	Fe-MOF/CMC	Enzyme induction	2.5-fold	48 h	Increased SOD activity	[21]
Paddy water	ZIF-8	Zn leaching	<0.5 mg L ⁻¹	30 days	Below WHO limits	[8]
Groundwater	UIO-66	Particle mobility	<5.0 cm	14 days	High soil filtration	[49]
Soil microbiome	MIL-100(Fe)	Evenness index	15.0% increase	90 days	Restoration of diversity	[29]
Wheat shoot	ZIF-8	Chlorophyll content	18.0% increase	25 days	Enhanced greenness	[56]
Soil aggregates	MOF-Biochar composite	Water holding capacity	2.95% per % SOC	—	Improved water storage	[29]
Leaf surface	PDA@ZIF-8	Adhesion work	45.2 mN m ⁻¹	Static	Sustained foliar contact	[50]
Grain starch	Fe-MOF (OPA)	Ferritin level	22.0% increase	At maturity	Nutritional fortification	[7]

probiotic inoculants decrease root malondialdehyde by 42.00% and proline accumulation by 39.13% and increase the number of grains by 145.05% in wheat grown under saline-alkali stress [56]. These reactions show improved nutrient uptake and decreased oxidative and osmotic stress.

The biofortification potential of such materials is also enhanced by elemental composition. Fe-MOFs with a composition of about 14.69% Fe, 14.48% P, and 6.03% N offer various nutrients in a single crystalline structure [26]. These elements are released by progressive mineralization at rates that are regulated by soil moisture, temperature and microbial activity. This controlled discharge minimizes the chances of overloading of metals and sustaining nutrient pools in the plants. Besides the provision of nutrients, the released metals become part of the plant metabolic pathways. Fe and Zn are cofactors of redox-active enzymes, play a role in chlorophyll biosynthesis, and in antioxidant defense mechanisms. These biochemical functions relate the release of nutrients by MOFs to quantifiable biomass growth and stress resistance [77]. In such circumstances, MOFs serve as temporally controlled nutrient reservoirs which increase crop productivity and reduce environmental loading [78].

Analysis of the reviewed literature shows clear relationships between MOF structural properties and environmental performance, which is a key challenge that needs to be tackled by long-term field trials and pilot-scale studies. The high surface area and hierarchical porosity increase the adsorption capacity, and the functionalized linkers increase the selectivity towards the pollutants. Moreover, the use of biodegradable linkers and environmentally friendly metal nodes helps to minimize environmental hazards. The observations allow for the development of initial design guidelines for future generation of agricultural MOFs, highlighting the need to combine high sorption efficiency, controlled degradability and biocompatibility. MOFs can enhance resource-use efficiency by releasing nutrients from the framework in a controlled manner, which minimizes nutrient losses through leaching and increases nutrient availability to plants, compared to conventional soluble fertilizers. While current studies show improved nutrient utilization and recovery, there is a lack of standardized quantitative comparisons of nutrient recovery efficiency and circular recycling pathways under field conditions, which needs to be investigated further.

5.2 Impacts on non-target soil macrofauna

Since MOFs are currently being used as amendments to functional soils but are currently being converted into mineralized residues, their ecological impacts need to be considered at the level of entire-soil communities. Ecotoxicological assessment offers a quantitative method of establishing the presence of degradation products on soil fauna, microbial diversity, or trophic interactions [76]. *Eisenia fetida* is a common earthworm that is used as a bioindicator due to its sensitivity to xenobiotic compounds and its central position in organic matter turnover and nutrient cycling. MIL@A@D@C nanocomposites have high biosafety margins in acute toxicity assays. Adult *E. fetida* have a survival rate near 100% at surface loadings of $251.6 \mu\text{g cm}^{-1}$ and above in contact tests that last 48 h [21]. There are no abnormal reactions, such as coiling, paralysis, or epidermal damage. These findings suggest little toxicity of dermal or ingestion. It seems that encapsulation in the MOF matrix decreases the bioavailability of active ingredients in the short term, thus limiting the exposure of organisms, as shown in Fig. 7.

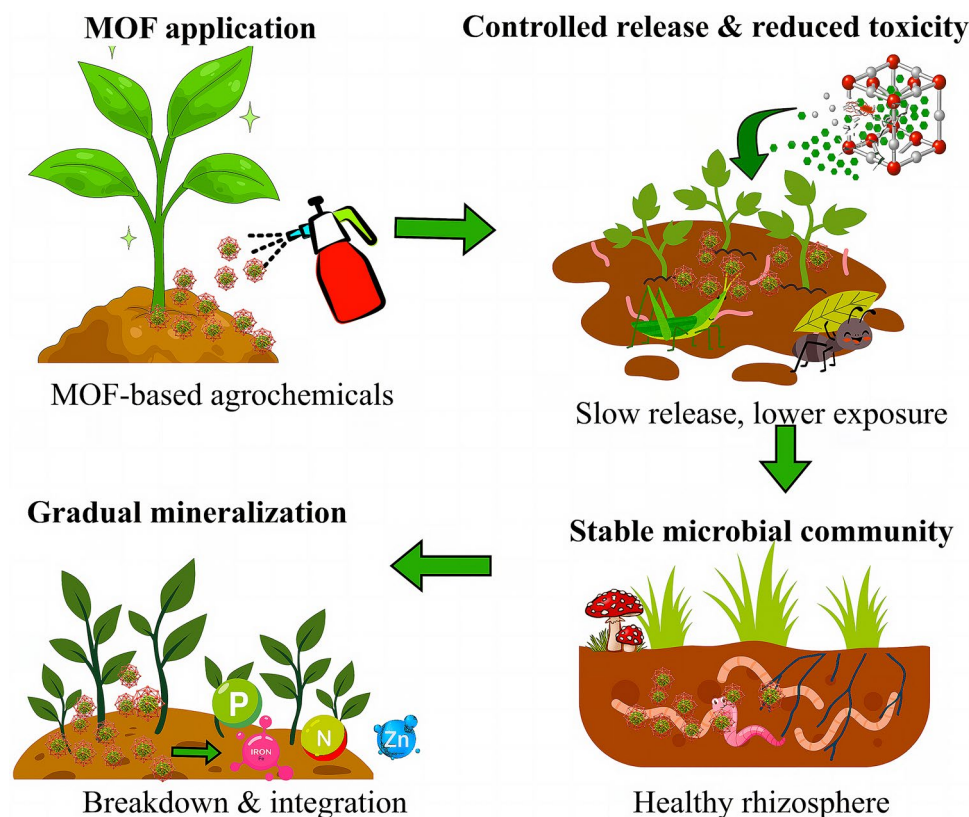


Fig. 7 The figure illustrates the environmental advantages of MOF-based agrochemicals in the soil systems. The regulated release, slow conversion, and biological compatibility of MOF with soil help in enhanced nutrient cycling and sustainable crop yields

Reduced acute toxicity compared to traditional pesticide preparations has been observed in several systems. FeMOF-pectin composite loaded with pyraclostrobin results in an eight-fold decrease in acute toxicity to zebrafish relative to commercial suspension concentrates [48]. Effects on positive rhizosphere microorganisms are also significant. Cu-BTC is a copper-based material that improves the growth of faba beans, boosting the dry biomass by 40.42% and leaf surface area by 46.22% (Mahmoud, 2025). Nevertheless, due to the accumulation of Cu^{2+} , the dosage needs to be optimized to avoid sensitive microbial taxa. Ecological compatibility is also shown by microbial community analyses. The native genera, such as biofilm-associated *Sphingobium* and *Lactococcus*, do not change or decrease in abundance after ZIF-8 co-assembly treatments. There is an increase in absolute population and a change in soil chemistry, such as a drop in pH by 0.63 units and a drop in salt content by 0.11 g kg^{-1} [56]. These results indicate that the degradation products are added to the rhizosphere matrix without interfering with functional microbial diversity. During the last phase of the MOF life cycle, crystalline structural characteristics are lost, metal ions are segregated into exchangeable or mineral-bound fractions, and organic connectors are integrated into the pools of soil organic matter. The high macrofaunal survival, low aquatic toxicity, and consistent microbial community structure are all indicative of the fact that, at controlled rates, MOF-based systems break down to components that can be incorporated into the existing soil biogeochemical processes [26, 56].

While there are reports of low acute toxicity for several MOFs, the majority of the evidence is from short-term laboratory testing. Long-term exposure may lead to the gradual release and accumulation of metal ions (e.g., Fe, Zn, and Cu) in soil, which could affect soil organisms. Furthermore, sublethal effects on growth, reproduction and behavior are not well known. Therefore, long-term field studies are needed to confirm the environmental safety of MOFs.

5.3 Aqueous mobility and groundwater risk

The long-term environmental safety of MOFs and their degradation products is defined by their vertical mobility. Downward water flux is caused by rainfall and irrigation, and remaining nanoparticles enter groundwater aquifers. That is why the mobility of MOF-derived species in soil should be assessed quantitatively [26, 49]. The primary factors that regulate the leaching of MOF-derived metal nodes are the soil pH and cation exchange capacity. These parameters regulate the solubility of the ions, adsorption and retention in the soil profile [79]. Despite this pH sensitivity, there is a limitation of vertical transport by soil buffering mechanisms. As an example, Fe^{3+} released by structures such as MIL-101(Fe) or OPA-MOF is rapidly hydrolyzed to insoluble iron oxyhydroxide or binds to clay minerals and soil organic matter. These sorption processes cause iron to accumulate in the upper 0–20 cm of soil, reducing the risk of iron being washed to groundwater [7, 26].

Besides dissolved ions, the potential mobility of intact or partially degraded MOF nanoparticles has also been studied. Colloidal transport relies on surface charge (zeta potential) and interaction with natural organic matter. Hydrophobic functionalization of surfaces with stearic acid (SL) or polydopamine (PDA) enhances the particles' aggregation. Enhanced aggregation improves physical straining and retention in soil pore networks. Sustained cargo release was found in UiO-66- NH_2 systems with SL modification over 11–52 h, yet intact nanoparticles were still mostly restricted to the topsoil because of hetero-aggregation with mineral surfaces [49, 50]. Framework composition also has an impact on mobility. MOFs based on Fe- and Zn-tend to mineralize comparatively fast in soil. Conversely, zirconium-based nodes based on UiO-66 exhibit a high chemical stability and hydrolysis resistance. Nevertheless, zirconium has a high affinity to phosphate, and thus, zirconium forms insoluble zirconium–phosphate complexes in soil. This reaction fixes zirconium species and restricts the movement of groundwater [17]. Inductively coupled plasma mass spectrometry is usually used in long-term monitoring of leachate to identify trace metal concentrations. The existing field and simulation data show that, at the recommended agricultural application rates, the concentrations of MOF-derived metals in groundwater are significantly lower than the drinking water guideline values, such as $\text{Zn} < 3 \text{ mg} \cdot \text{L}^{-1}$ and $\text{Fe} < 0.3 \text{ mg} \cdot \text{L}^{-1}$ [8, 26]. These results indicate that the soil chemical and physical retention processes restrict vertical migration of MOF-derived components under controlled use conditions.

6 Safety, regulation, and soil health benchmarks

As MOFs begin to be used in agriculture, it is important to ensure that they are safe for soil, crops, and human health. The safe-by-design approach focuses on selecting materials and designing structures that can safely break down in the soil without harming the environment [80].

6.1 Defining “safe-by-design” agricultural MOFs

Safe-by-design approach takes into account the compatibility with the environment during the synthesis of MOFs. This strategy combines the choice of metals, structural lability, and end-of-life transformation into a single design principle [8]. One element of this strategy is the application of metal nodes, which are associated with vital plant micronutrients. Fe, Zn, Mg, and Ca are preferred because, upon degradation of the framework, they contribute to the existing nutrient pools in the soil rather than introducing non-native elements [26].

The important design parameter is the synthetic methodology that uses water or ethanol instead of high-boiling aprotic solvents like *N,N*-dimethylformamide, which will decrease the residual amount of solvent and enhance the sustainability process, as illustrated in Fig. 8. In the encapsulation of AZOX in $\text{NH}_2\text{-MIL-101(Fe)}$, the highest loading capacity of 6.73% was obtained using ethanol-based synthesis, which shows that the functional performance is not impaired by solvent substitution [21]. Structural lability may also be adjusted to agricultural time scales. Experiments conducted in the field indicate that Fe-MOFs lose 50.9% of their mass in six months of wheat growth [16]. Verification of end-of-life transformation is required to confirm that the designed degradability

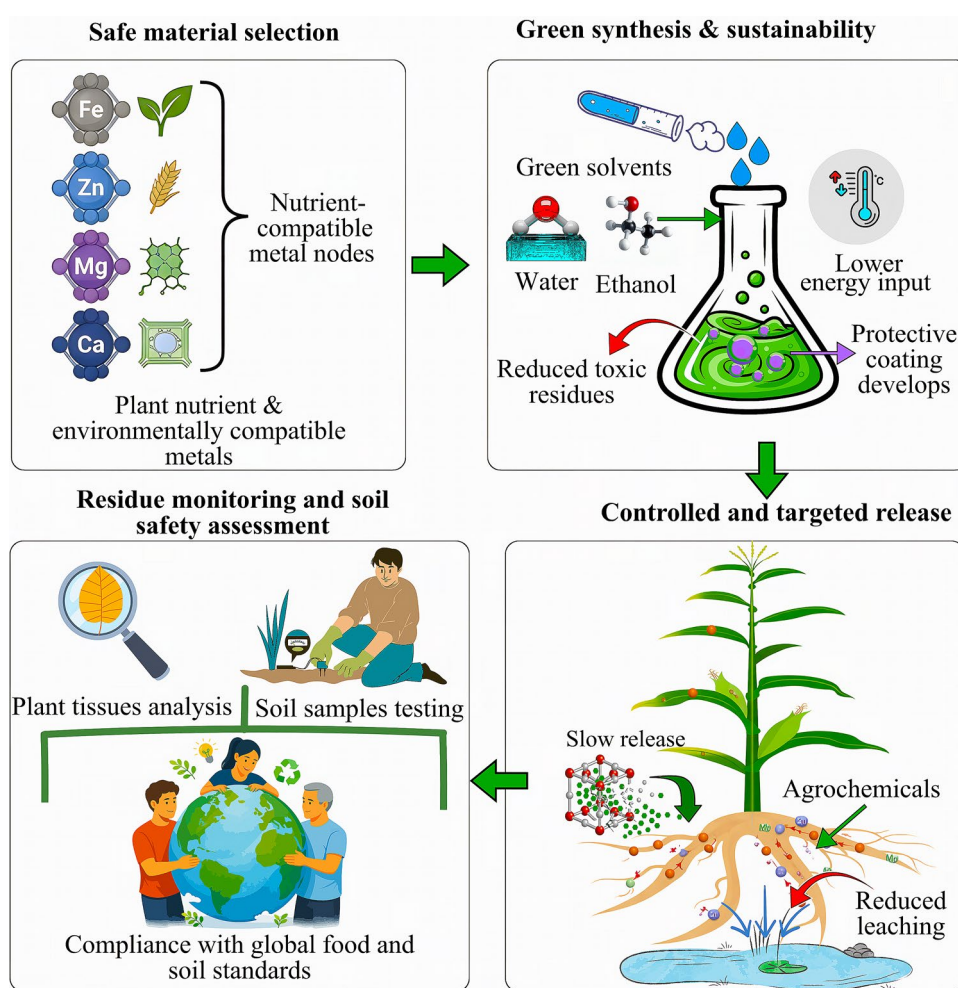


Fig. 8 Theoretical overview of a safe-by-design approach that combines the choice of materials, green synthesis, controlled delivery, and post-use safety assessment to promote environmental compatibility and regulatory compliance of the MOF-based system

results in environmentally compatible products. In Fe-based systems, progressive loss of crystallinity is observed through attenuation and disappearance of characteristic PXRD reflections, including the (110) and (111) planes near 28° and 31° , together with weakening of FTIR carboxylate stretching bands at 1650 and 1430 cm^{-1} . These changes correspond to conversion into amorphous iron oxyhydroxides such as ferrihydrite [26, 81].

6.2 Compliance with global food and soil standards

The commercialization of MOF-based technologies in laboratory research and agricultural practice must adhere to international regulatory standards, as shown in Fig. 8. This guarantees that crop protection and nutrient delivery functions do not produce unwanted residues in soils or food products [82]. Regulatory assessment is based on three interrelated criteria, such as elemental loading, active ingredient residues and biodegradability [83].

In the guidelines established by the Food and Agriculture Organization and the World Health Organization through the Joint FAO/WHO Meeting on Pesticide Residues, agricultural applications should not exceed the set maximum residue limits (MRLs) of soils and edible plant tissues. In the case of micronutrient-based systems like ZIF-8 and Cu-BTC, the calculations of dosage are aimed at keeping soil metal levels below generally accepted precautionary levels, usually less than 300 mg kg^{-1} and 100 mg kg^{-1} of Zn and Cu, respectively, in agricultural soils [58]. Experiments in the field have shown that ions released under controlled conditions in MOF matrices can be kept at concentrations significantly lower than these limits. In hybrid systems of pesticides and fertilizers, Zn^{2+} is emitted in low amounts that ensure that plant tissue does not exceed 100 mg kg^{-1} , thus, satisfying food-safety requirements and micronutrient sufficiency [84, 85]. In the case of nanoparticles used for pesticide delivery, regulatory compliance focuses on the minimization of terminal residues in crops produced. Acetamiprid@UiO-66-NH₂-CMC encapsulation systems can load up to 90.79% and decrease the total environmental input by reducing volatilization, spray drift, and runoff [19]. The same results are obtained with Boscalid@ZIF-67 and IMI@Fe₃O₄-PDA-UiO-66 (15.87% loading). These systems offer sustained-release profiles similar to commercial water-dispersible granules, and reduce final crop residues by 30–50% compared to conventional foliar treatments [46, 47]. These cuts enhance the difference between measured residues and statutory MRLs.

Existing research indicates that the concentrations of metals in the drinking water from MOFs are not expected to exceed drinking water guidelines under controlled experimental conditions, but the results are from a relatively small number of studies with short monitoring periods. The vertical mobility of the degradation products of dissolved and colloidal MOFs in realistic irrigation, rainfall and seasonal field conditions is still not well understood. Future studies should involve multi-season field studies to assess the transport of metals in the form of colloids, groundwater movement, and the overall environmental fate of metals released from MOFs under various soil and climatic conditions.

The analysis of post-harvest plant tissues gives ultimate confirmation of compliance with the regional food-safety regulations, including the control of the European Food Safety Authority. In the case of nitrogen-delivery materials like N-P-MIL-156-H (32.3% total nutrient content), regulated release of nutrients prevents high levels of nitrates, keeping the concentrations in leafy vegetables within the $2,000\text{--}5,000\text{ mg kg}^{-1}$ range of

Commission Regulation (EU) No 1258/2011. GR-MOF-27 also promotes the uptake of Mg and P, with a 10.5% and 13.1% increase in shoot and root dry mass, respectively, without reaching nutritional safety limits or altering organoleptic characteristics [27]. Ecotoxicological assessment also promotes regulatory acceptance. CLO@ZIF-8 raises the acute toxicity threshold of honeybees by at least 120-fold over the free active compound, and PYR@FeMOF-pectin decreases the toxicity of zebrafish by eightfold [48, 71]. Similar rises in EC_{50} values compared to non-encapsulated formulations are signs of increased ecological safety margins and decreased risk to non-target organisms [8, 21]. Combined, these results indicate that MOF-based agricultural inputs designed with a compliance-based design framework can comply with international residue standards, be subject to significant in-season mineralization, and be able to sustain crop concentrations within legal safety limits. This combined strategy helps in protecting the environment and the food chain.

6.3 Long-term soil health indicators

The recovery of soil biological integrity and physical structure following remediation is used as a measure of the long-term performance of MOF-based soil amendments. Evaluation thus goes beyond the elimination of contaminants and is aimed at stabilizing the ecology, so that the detoxification process is accompanied by the restoration of natural soil processes [28]. A primary indicator of recovery is the restoration of microbial community structure and functional diversity after alleviation of stress caused by pollutants. Selective pressure on the soil microbiome reduces when MOFs bind heavy metals such as Cd and Pb and break down the persistent organic pollutants like PNP and atrazine. This enables the ecologically important, but hitherto suppressed, taxa such as *Acidobacteriota* and *Actinomycetales* to re-colonize polluted microhabitats [26, 29]. Experiments in the field and mesocosm indicate that frameworks based on iron experience significant structural dissociation in a single growing season (usually 90–180 days). This reduces the residual metal bioavailability and does not increase the overall Fe concentrations to agronomically acceptable levels [26, 29].

The degradation of iron-based MOFs also activates oxalotrophic bacteria, whose oxalate linkers are released and metabolized by the oxalate–carbonate pathway. This route facilitates mineralization of the remaining framework elements and increases the stability of microbial networks [7, 26]. Following remediation, soils often have a higher level of microbial richness and evenness. The high-throughput 16S rRNA gene sequencing shows that the genes related to nitrogen fixation and phosphorus solubilization revert to the near-baseline abundance after MOF treatment, which proves the restoration of self-regulating nutrient cycling processes [8, 21].

Besides microbiological restoration, MOF mineralization has an effect on the physical properties of soil. Framework degradation emits Fe, Al and Ca ions, along with organic ligands which act as binding agents. These elements enhance the development of stable micro-aggregates in the topsoil layer of 0–20 cm. Secondary minerals, such as iron oxyhydroxides formed based on the degraded structures, also contribute to the strengthening of particles and their connectivity [16, 29]. WHC is enhanced by the incorporation of mineralized linkers into the SOC pool, and by the hierarchical pore structure of MOF-biochar composites like WSB-MIL-100. MOF amendments enhance the available water capacity in sandy soils by 1.16–2.95% per unit mass addition of organic carbon.

The formation of mesopores with a 0.5–50 μm size also increases moisture retention and hydraulic conductivity [49]. Responsive MOF systems with sustained-release behavior lasting up to 11–52 h allow gradual integration of the mineral and avoid sharp physico-chemical disruptions, which promote progressive structural stabilization [49].

Future research should aim for an integrated soil regeneration indicator framework to allow for a standardized assessment of the remediation achieved by MOF. We suggest the concept of a MOF Soil Regeneration Index (MSRI) that combines four groups of indicators: (i) microbial resilience indicators (community diversity, functional gene recovery), (ii) nutrient cycling restoration (N fixation, P solubilization, SOC stabilization), (iii) contaminant immobilization persistence (residual bioavailability and leaching risk), and (iv) soil structural recovery (aggregate stability, porosity, water retention). A multi-dimensional indicator system would allow cross-study comparison, long-term field monitoring and life-cycle assessment of MOF technologies. Creating common soil health metrics will be key to moving soil health lab results to regulatory approval and adoption by farmers.

7 Future directions: autonomous and circular remediation

The future of agricultural MOFs will move from material-based remediation and passive delivery to autonomous and circular remediation platforms that can sense, respond and degrade the environment. Future MOFs can be designed to be self-regulating rhizosphere platforms that actively regulate soil health instead of just removing contaminants, as shown in Figure 9 (Bharti et al., 2024). This development is a move towards closed-loop soil remediation, where the capture, remediation and recovery of contaminants take place in parallel. These intelligent remediation systems are in line with the new paradigm in agriculture, which is circular agriculture, where external chemical inputs are reduced and in-situ regeneration of soil functionality, nutrient cycling and ecosystem resilience are optimized.

7.1 Smart sensors and theranostic MOFs

The concept of theranostic MOFs in agriculture integrates contaminants detection localized remediation. These systems are evaluated using quantitative measurements and comprise detection limits, response time, release kinetics, and adsorption capacity [86]. Unlike the conventional passive sorbents, the advanced MOFs are designed as dual-functional platforms that identify the contaminants at low concentrations and with high sequestration capacity, as shown in Fig. 9. Future research should focus on developing autonomous MOF systems that can dynamically respond to fluctuations in contaminant concentration, soil pH, redox potential, and microbial activity without external intervention. Beyond laboratory-scale demonstrations, future research should prioritize large-scale field validation under diverse soil, climatic, and cropping conditions to evaluate the robustness, reproducibility, and long-term performance of theranostic MOF platforms. The development of standardized field evaluation protocols and long-term environmental monitoring frameworks may also facilitate cross-study comparison and support future regulatory assessment.

Luminescent MOFs (LMOFs) have detection limits of the low parts-per-billion of metal ions and agrochemicals of environmental interest. Frameworks based on zirconium functionalized with fluorescent ligands have fluorescence quenching efficiencies

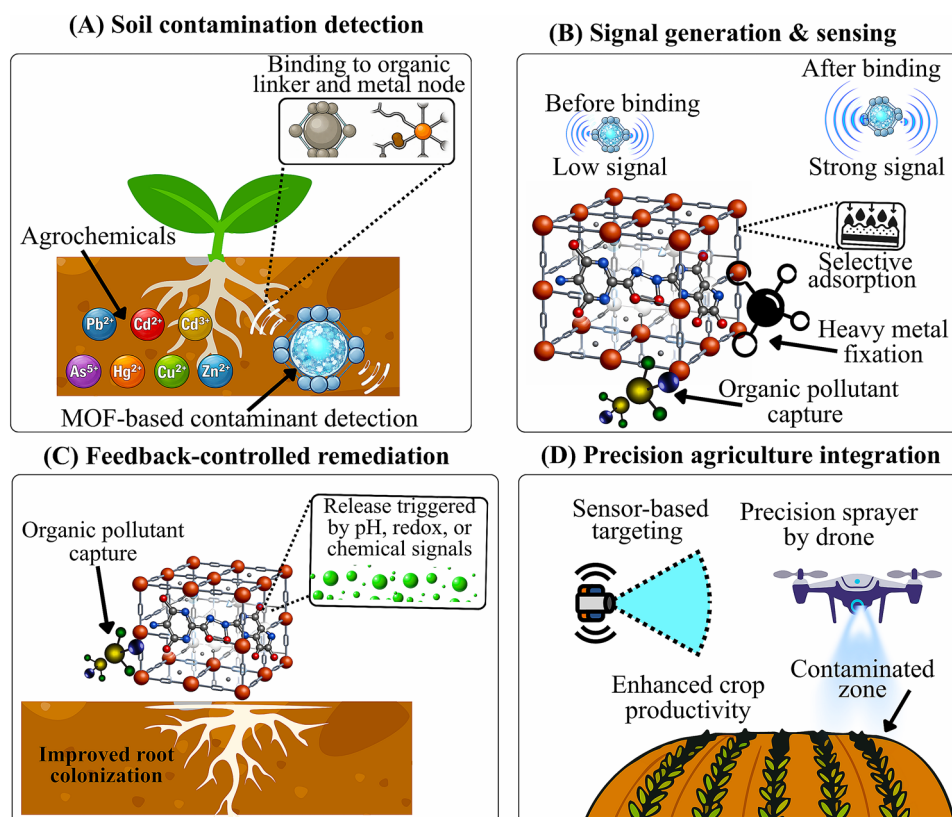


Fig. 9 Overview of MOF-based procedures of (A) contaminant detection and (B) signal amplification to selective capture of pollutants, (C) stimulus-responsive release, and (D) precision application of the sensor, resulting in improved soil health and crop productivity

of 70–90% in the presence of analytes such as AsO_4^- or Cd^{2+} at concentrations below 10 ppb. This type of reaction enables optical detection and adsorption to take place simultaneously in microporous spaces (<2 nm pore diameter) [29]. The reported Stern Volmer quenching constants (K_{sv}) are typically 10^4 – 10^5 M^{-1} heavy metal sensing, meaning that the analyte and the framework interact strongly, and response times are in the range of a few seconds to several minutes. Signal enhancement factors may also exceed 10^6 cases. LMOFs are used with surface-enhanced Raman scattering (SERS) substrates, which also reduces practicable detection limits in the field.

Post-synthetic modification increases the density of coordination and recognition sites to approximately 1 – 3 mmol g^{-1} . This increased the sensitivity of the analysis and the adsorption capacity. Fe- and Zr-based MOFs typically possess Pb^{2+} and Cd^{2+} adsorption capacities of 100 – 300 mg g^{-1} and maintain fluorescence responsive [47, 50]. Theranostic MOFs can also be controlled by feedback release, in which the binding of contaminants regulates the release of remedial agents. Controlled-release systems are Fe-based systems that release nutrients or ligands within 240–1152 h (10–48 days) based on the soil pH and ionic composition [26]. In competitive ligand-exchange systems, release can be programmed to take place when the free Pb^{2+} concentrations in the soil pore water become 0.1 – 0.3 mg L^{-1} , which are phytotoxicity-relevant concentrations. GR-MOF-20 and other conceptual Agro MOFs are designed to maintain Pb^{2+} adsorption capacity of more than 150 mg g^{-1} and restrict the release of chelators to local contamination areas, thereby reducing the consumption of unnecessary chemicals [26, 56]. Future studies

should therefore prioritize multi-functional “closed-loop” MOF architectures capable of simultaneous sensing, targeted remediation, nutrient recycling, and environmentally synchronized degradation to support circular and precision agriculture systems.

7.2 AI-assisted selection of circular linkers

The development of a zero-waste agricultural system requires materials whose functional activity is directly linked to controlled and predictable environmental degradation. Artificial intelligence (AI) assisted high-throughput screening provides a systematic method to select MOF components that maintain structural stability during application but undergo programmed dissipation after use [87]. Computational toxicology also limits the choice of material before manufacturing. In silico systems assess the predicted bioaccumulation potential, redox reactivity and cytotoxicity. It is only structures that are projected to retain $\geq 79\%$ of microbial viability following seven days in nutrient-limited environments that are advanced to be developed [21]. Climatic variables are also used in integrated predictive models to estimate environmental persistence. Indicatively, regression models replicate the experimentally determined growth of monthly MOF degradation between 4.2% and 13.8% at 4.7 °C and 19.8 °C, respectively, with $R^2 = 0.99$, a direct correlation between temperature profiles and structural half-life [16]. Future studies should expand these predictive systems by integrating soil microbial diversity, rhizosphere chemistry, and seasonal climatic variability into AI models to improve field-level accuracy and long-term environmental forecasting. By optimizing the structural stability, degradability, remediation efficiency, and environmental compatibility simultaneously, machine learning and AI-assisted material discovery can further expedite the identification of biodegradable MOFs and hybrid MOF–biochar composites. Integration of high-throughput computational screening with experimental validation could substantially reduce material development time while improving design reliability.

Deep learning techniques assist in structural optimization by matching the synthesis parameters to porosity properties. The algorithm-guided tuning of the WSB@MIL-100 composite reached a certain surface area of $419 \text{ m}^2 \text{ g}^{-1}$, which is significantly greater than that of similar rice straw/MIL-100(Fe) hybrids ($133.9 \text{ m}^2 \text{ g}^{-1}$) [29]. The higher the surface area, the better the kinetics of adsorption and immobilization of contaminants, thus the less material is needed per unit area of soil. Stimuli-responsive release behavior is also controlled using AI models. Multivariate algorithms combine soil pH, pathogen activity and molecular diffusion coefficients to establish activation thresholds that reflect biologically induced microenvironmental changes, including localized acidification during infection by *Magnaporthe oryzae*. This targeted response is linked to an eightfold lower acute toxicity to zebrafish in pyraclostrobin-loaded Fe-MOF systems than in non-encapsulated formulations, which implies a lower off-target exposure [48]. Predictive algorithms have the potential to estimate adsorption performance under field-relevant geochemical conditions, although further experimental validation is required. MOF composition and pore architecture have been optimized to soils with up to $142 \text{ mmol kg}^{-1} \text{ Cu}^{2+}$ and be effective at immobilizing the soils without compromising compatibility with native microbial communities and mineral matrices [29]. Despite these advances, successful translation into agricultural practice will also require techno-economic analysis, life-cycle assessment, scalable green synthesis, and regulatory approval to evaluate manufacturing costs, environmental sustainability, and practical feasibility.

Addressing these commercialization challenges may accelerate the responsible deployment of AI-designed MOF technologies for precision and circular agriculture.

At the ecosystem level, AI-based environmental risk models estimated microbial compatibility and toxicity levels of MOF materials. Simulations of laboratories suggest that microbial respiration in soil does not drop below 85–90% of baseline activity following exposure to Fe-MOF concentrations less than 250 mg kg⁻¹ soil, whereas higher concentrations start to decrease microbial biomass carbon by 12–18% over 30 days of incubation [88]. AI-driven methods could complement traditional empirical optimization by incorporating degradation kinetics, ecotoxicological data, climate factors, and sorption thermodynamics in computational models. Moving forward, there is a need for the creation of explainable AI frameworks and standardized environmental datasets to enhance the transparency, reproducibility, and potential regulatory acceptance of AI-assisted MOF technologies. As these methods develop and are further validated, they can help to achieve more sustainable agronomic performance, precision remediation and better nutrient management.

7.3 Scaling strategies for large-area deployment

The transition of MOF-based remediation technologies from small laboratory production to large-scale field use is an important step for their practical application [89]. At this larger scale, success depends not only on how well the material works but also on several other factors, including the availability of raw materials, efficiency of synthesis, behavior in agricultural soils, and how the material breaks down in the environment at the end of its use [40]. Agronomic return is becoming more and more considered in economic models. GR-MOF-27 boosts shoot, root and total dry biomass by 10.5%, 11.0%, and 13.1%, respectively, relative to conventional MgSO₄ fertilization (Morales-Camara, 2025). These findings indicate the potential to offset synthesis costs; however, confirmation under large-scale field conditions is still required.

Adaptability to the current agricultural infrastructure also promotes adoption. The seed-coating system, like IPC@UiO-66-TA-Zn²⁺ is composed of monodisperse spherical particles (~296 nm) which can be easily used with the traditional mechanical seeders, thus avoiding the need to modify equipment. Efficacies of reported pathogen control are 84.09–93.10%, which is higher than conventional suspension concentrates. The emergence rates of seedlings are 88–92.67%, which means that the germination performance is maintained [8]. This is an operational compatibility, which enables it to integrate into the existing agronomic practices. Commercialization is likely to be enhanced by regulatory developments. With more stringent rules and restrictions on heavy metal soil concentration, Fe-based MOFs offer a degradable alternative to the non-biodegradable chelating agents. Slow transformation into stable mineral phases facilitates adherence to the regulations of trace-metal mobility and environmental persistence [7]. Taken together, these developments suggest the emergence of a data-driven, self-optimizing remediation paradigm in which MOFs evolve from static materials into adaptive environmental technologies. Future research should therefore prioritize the integration of sensing networks, long-term field trials, and regulatory frameworks to enable the transition of MOF technologies from experimental demonstrations to resilient agricultural infrastructure.

8 Conclusion

This review underscores the potential of MOFs as a revolutionary material for contaminated soil remediation and regeneration in agriculture. MOFs are engineered to perform multiple functions, such as pollutant adsorption, controlled delivery of agrochemicals, microbial compatibility, and environmentally synchronized degradation, transforming the concept of remediation from a one-function intervention to a multifunctional soil management strategy. MOFs allow for the removal of contaminants without compromising soil structure or biological activity, unlike traditional approaches that can damage soil structure and biodiversity. One of the major benefits of MOFs that are environmentally friendly is their life-cycle functionality. Once pollutants are captured or agrochemicals released, many MOFs degrade slowly over time via microbial activity, leaving behind stable mineral products and releasing plant nutrients. This property enables MOFs to be used as regenerative materials that can integrate pollution control with nutrient recycling, soil fertility improvement and ecosystem restoration, which will facilitate a transition to circular soil management. Future developments in stimuli-responsive release, smart sensing, and AI-driven material design may further advance MOF technologies, although their practical performance requires validation under diverse field conditions. But, field validation, scalable synthesis, life-cycle assessment and regulatory frameworks are still essential for real-world implementation. In summary, the synergy of nanotechnology, soil science, and sustainable agriculture suggests a circular agro-nanotechnology approach where innovative materials contribute to pollution reduction, nutrient recovery, and sustainable food production. Interdisciplinary research will continue to be crucial to bring innovations from the lab to the field and to the farm.

Author contributions

A1: Conceptualization, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft preparation. A2: Validation, Writing – review and editing. A3: Validation, Writing – review and editing. A4: Supervision, Project administration, Validation, Writing – review and editing. All authors contributed to the development of the manuscript, critically reviewed the content, and approved the final version for publication.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethical approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 19 June 2026 / Revised: 28 July 2026 / Accepted: 20 August 2026

Published online: 01 September 2026

References

1. Kopittke PM, Menzies NW, Wang P, McKenna BA, Lombi E. Soil and the intensification of agriculture for global food security. *Environ Int.* 2019;132:105078. <https://doi.org/10.1016/j.envint.2019.105078>.
2. Petruzzelli G, Pezzarossa B, Pedron F. The fate of chemical contaminants in soil with a view to potential risk to human health: a review. *Environments.* 2025;12:183. <https://doi.org/10.3390/environments12060183>.

3. Sarkar S, Jaswal A, Singh A. Sources of inorganic nonmetallic contaminants (synthetic fertilizers, pesticides) in agricultural soil and their impacts on the adjacent ecosystems. In: *Bioremediation of emerging contaminants from soils*. Elsevier; 2024. p. 135–61. <https://doi.org/10.1016/B978-0-443-13993-2.00007-4>.
4. Hossain ME, Shahrukh S, Hossain SA. Chemical fertilizers and pesticides: impacts on soil degradation, groundwater, and human health in Bangladesh. In: *Environmental degradation: challenges and strategies for mitigation*. Springer; 2022. p. 63–92. https://doi.org/10.1007/978-3-030-95542-7_4.
5. Omo-Okoro P, Ofori P, Amalapidman V, Dadrasnia A, Abbey L, Emenike C. Soil pollution and its interrelation with interfacial chemistry. *Molecules*. 2025;30:2636. <https://doi.org/10.3390/molecules30122636>.
6. Jarvis N. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *Eur J Soil Sci*. 2020;71(3):279–302. <https://doi.org/10.1111/ejss.12973>.
7. Anstoetz M, Rose TJ, Clark MW, Yee LH, Raymond CA, Vancov T. Novel applications for oxalate-phosphate-amine metal-organic-frameworks (OPA-MOFs): can an iron-based OPA-MOF be used as slow-release fertilizer? *PLoS One*. 2015;10:e0144169. <https://doi.org/10.1371/journal.pone.0144169>.
8. Zhang Q, Shi X, Gao T, Xing Y, Jin H, Hao J, et al. Precision management of fusarium fujikuroi in rice through seed coating with an enhanced nanopesticide using a tannic acid-ZnII formulation. *J Nanobiotechnol*. 2024;22(1):717. <https://doi.org/10.1186/s12951-024-02938-y>.
9. Grifoni M, Franchi E, Fusini D, Vocciante M, Barbaferri M, Pedron F, et al. Soil remediation: towards a resilient and adaptive approach to deal with the ever-changing environmental challenges. *Environments*. 2022;9:18. <https://doi.org/10.3390/environments9020018>.
10. Ghisman V, Muresan AC, Bogatu NL, Herbei EE, Buruiana DL. Recent advances in the remediation of degraded and contaminated soils: a review of sustainable and applied strategies. *Agronomy*. 2025;15:1920. <https://doi.org/10.3390/agronomy15081920>.
11. Cui W, Li X, Duan W, Xie M, Dong X. Heavy metal stabilization remediation in polluted soils with stabilizing materials: a review. *Environ Geochem Health*. 2023;45:4127–63. <https://doi.org/10.1007/s10653-023-01522-x>.
12. Negi M, Sharma S, Singh M. Plant-driven soil dynamics: exploring the role of roots, fungi and soil traffic in shaping the soil structure. *Commun Soil Sci Plant Anal*. 2025;56:2727–46. <https://doi.org/10.1080/00103624.2025.2524075>.
13. Bindraban PS, Dimkpa CO, Pandey R. Exploring phosphorus fertilizers and fertilization strategies for improved human and environmental health. *Biol Fertil Soils*. 2020;56:299–317. <https://doi.org/10.1007/s00374-019-01430-2>.
14. Khan SH. Green nanotechnology for the environment and sustainable development. *Green materials for wastewater treatment*. Springer; 2019. p. 13–46. https://doi.org/10.1007/978-3-030-17724-9_2.
15. Pokrajac L, Abbas A, Chrzanowski W, Dias GM, Eggleton BJ, Maguire S, et al. Nanotechnology for a sustainable future: addressing global challenges with the international network4sustainable nanotechnology. ACS Publications; 2021. <https://doi.org/10.1021/acs.nano.1c10919>.
16. Wu K, Du C, Ma F, Shen Y, Liang D, Zhou J. Degradation of metal-organic framework materials as controlled-release fertilizers in crop fields. *Polymers*. 2019;11:947. <https://doi.org/10.3390/polym11060947>.
17. Mahmoud LA, Telford R, Livesey TC, Katsikogianni M, Kelly AL, Terry LR, et al. Zirconium-based MOFs and their biodegradable polymer composites for controlled and sustainable delivery of herbicides. *ACS Appl Bio Mater*. 2022;5:3972–81. <https://doi.org/10.1021/acsabm.2c00499>.
18. Gatou M-A, Vagena I-A, Lagopati N, Pippa N, Gazouli M, Pavlatou EA. Functional MOF-based materials for environmental and biomedical applications: a critical review. *Nanomaterials*. 2023;13(15):2224. <https://doi.org/10.3390/nano13152224>.
19. Song S, Wan M, Feng W, Tian Y, Jiang X, Luo Y, et al. Environmentally friendly Zr-based MOF for pesticide delivery: ultrahigh loading capacity, pH-responsive release, improved leaf affinity, and enhanced antipest activity. *Langmuir*. 2022;38(35):10867–74. <https://doi.org/10.1021/acs.langmuir.2c01556>.
20. Bi L, Shi L, Wang K, Li J, Tao X, Liu P, et al. In-depth analysis of the removal mechanisms behind neonicotinoid pesticides and bimetal ZIF derived-magnetic carbon. *Separat Purif Technol*. 2025;354:129292. <https://doi.org/10.1016/j.seppur.2024.129292>.
21. Li Y, Zhang T, Ding Y, Chen Y, Rui Y. Development of a pH/Cellulase dual-responsive Fe-MOFs delivery system for targeted control of maize fungal diseases. *Langmuir*. 2025;41(36):24307–18. <https://doi.org/10.1021/acs.langmuir.5c02196>.
22. Chen M, Hu Y, Zhou X, Xia L, Wu C, Chen Y, et al. Microbial therapeutics for cancer: emerging strategies and biomedical applications. *Chem Commun*. 2025;61:14531–64. <https://doi.org/10.1039/D5CC03618F>.
23. Savadkouhi Ghoudjanaki M, Bahadorani F, Hosseini M, Panta R, Karimi B. Metal-based categorization of recent metal organic frameworks with application in bone disease treatment. *ACS Biomater Sciamp; Eng*. 2025;11:6895–929. <https://doi.org/10.1021/acsbiomaterials.5c00830>.
24. Bahrani S, Hashemi SA, Mousavi SM, Azhdari R. Zinc-based metal-organic frameworks as nontoxic and biodegradable platforms for biomedical applications: review study. *Drug Metab Rev*. 2019;51:356–PNP77. <https://doi.org/10.1080/03602532.2019.1632887>.
25. Basak S, Bhattacharyya P, Lokhande PE, Chakrabarti S. MOF: a New Age Smart material as Nano carriers for fertilizers and pesticides. In: *Biological applications of nanoparticles*. Springer; 2023. p. 135–48. https://doi.org/10.1007/978-981-99-3629-8_8.
26. Wu K, Xu X, Ma F, Du C. Fe-based metal-organic frameworks for the controlled release of fertilizer nutrients. *ACS Omega*. 2022;7(40):35970–80. <https://doi.org/10.1021/acsomega.2c05093>.
27. Morales-Cámara S, Cardona-Carrascosa L, Salcedo-Abraira P, Rodríguez-Diéguez A, Rojas S. Controlled magnesium release and nutritional effect of a novel metal-organic framework on plants. *Cryst Growth Des*. 2025;25:4685–89. <https://doi.org/10.1021/acs.cgd.5c00080>.
28. Mohan B, Sharma HK, Ručman S, Singjai P, Pombeiro AJ. Contaminants removal and monitoring: the role of hybrid MOFs in agricultural advancement and soil amendment. *Adv Sustain Syst*. 2025;9:2400637. <https://doi.org/10.1002/adsu.202400637>.
29. Polyakov V, Bauer T, Kirichkov M, Butova V, Gritsai M, Minkina T, et al. MOF-biochar nanocomposite for sustainable remediation of contaminated soil. *Environ Sci Pollut Research Environ Sci Pollut Res*. 2025;32:5533–50. <https://doi.org/10.1007/s11356-025-36052-9>.
30. Ren Y, Wang G, Bai X, Su Y, Zhang Z, Han J. Research progress on remediation of organochlorine pesticide contamination in soil. *Environ Geochem Health*. 2024;46:25. <https://doi.org/10.1007/s10653-023-01797-0>.

31. Raffa CM, Chiampo F. Bioremediation of agricultural soils polluted with pesticides: a review. *Bioengineering*. 2021;8(7):92. <https://doi.org/10.3390/bioengineering8070092>.
32. Delgado A, Gómez JA. The soil: physical, chemical, and biological properties. *Principles of agronomy for sustainable agriculture*. Springer; 2024. p. 15–30. https://doi.org/10.1007/978-3-031-69150-8_2.
33. Patra K, Dey S, Solanki C, Sengupta A, Mittal VK. Harnessing advanced porous materials, covalent organic frameworks, and porous organic polymers as next-generation porous frameworks for targeted removal of emerging water contaminants. *ACS Appl Eng Mater*. 2025;3:1130–65. <https://doi.org/10.1021/acsaenm.5c00187>.
34. Emmanuel SS, Aniekezie FC, Adesibikan AA. Covalent organic frameworks and metal–organic frameworks for sustainable adsorptive removal/extraction of dirty dozen chemicals: a review. *J Chin Chem Soc*. 2024;71(9):978–1007. <https://doi.org/10.1002/jccs.202400203>.
35. Meng W, Tian Z, Yao P, Fang X, Wu T, Cheng J, et al. Preparation of a novel sustained-release system for pyrethroids by using metal-organic frameworks (MOFs) nanoparticle. *Colloids Surf Physicochem Eng Asp*. 2020;604:125266. <https://doi.org/10.1016/j.colsurfa.2020.125266>.
36. Abdelhameed RM, Taha M, Abdel-Gawad H, Hegazi B. Amino-functionalized Al-MIL-53 for dimethoate pesticide removal from wastewater and their intermolecular interactions. *J Mol Liq*. 2021;327:114852. <https://doi.org/10.1016/j.molliq.2020.114852>.
37. Chen H, Shan Y, Cao L, Zhao P, Cao C, Li F, et al. Enhanced fungicidal efficacy by co-delivery of azoxystrobin and diniconazole with cauliflower-like metal-organic frameworks NH₂-al-MIL-101. *Int J Mol Sci*. 2021;22:10412. <https://doi.org/10.3390/ijms221910412>.
38. Li Q, Wang L, Fu Y, Lin D, Hou M, Li X, et al. Transformation of soil organic matter subjected to environmental disturbance and preservation of organic matter bound to soil minerals: a review. *J Soils Sediments*. 2023;23(3):1485–500. <https://doi.org/10.1007/s11368-022-03381-y>.
39. Ukalska-Jaruga A, Beijger R, Smreczak B, Podlasiński M. Sorption of organic contaminants by stable organic matter fraction in soil. *Molecules*. 2023;28:429. <https://doi.org/10.3390/molecules28010429>.
40. Yang C, Xue Z, Wen J. Recent advances in MOF-based materials for remediation of heavy metals and organic pollutants: insights into performance, mechanisms, and future opportunities. *Sustainability*. 2023;15:6686. <https://doi.org/10.3390/su15086686>.
41. Tang S, Wang Y, He P, Wang Y, Wei G. Recent advances in metal-organic framework (MOF)-based composites for organic effluent remediation. *Materials*. 2024;17:2660. <https://doi.org/10.3390/ma17112660>.
42. Chaturvedi K, Singhwani A, Dhangar M, Mili M, Gorhae N, Naik A, et al. Bamboo for producing charcoal and biochar for versatile applications. *Biomass Convers Biorefin*. 2024;14:15159–85. <https://doi.org/10.1007/s13399-022-03715-3>.
43. Ramanathan V, Kh SG. Enhancing water stability of metal-organic frameworks via phosphonate monoester linkers. 2012. <https://doi.org/10.1021/ja306812r>.
44. Song L, Yuan Y, Wang Y, Zhang TC, He G, Yuan S. Cu-MOF-derived Cu nanoparticles decorated porous N-doped biochar for low-temperature H₂S desulfurization. *Fuel*. 2024;368:131682. <https://doi.org/10.1016/j.fuel.2024.131682>.
45. Zhang R, Lv G, Du C, Ma F, Liu S, Gan F, et al. Hierarchically porous metal-organic frameworks-based controlled-release fertilizer: improved nutrient loading and rice growth. *Agronomy*. 2025;15:2334. <https://doi.org/10.3390/agronomy15102334>.
46. Zhang X, Tang X, Zhao C, Yuan Z, Zhang D, Zhao H, et al. A pH-responsive MOF for site-specific delivery of fungicide to control citrus disease of botrytis cinerea. *Chem Eng J*. 2022;431:133351. <https://doi.org/10.1016/j.cej.2021.133351>.
47. Meng W, Gao Y, Tian Z, Xu W, Cheng J, Li S, et al. Fe₃O₄ magnetic cores coated with metal-organic framework shells as collectable composite nanoparticle vehicles for sustained release of the pesticide imidacloprid. *ACS Appl Nano Mater*. 2021;4:5864–70. <https://doi.org/10.1021/acsaenm.1c00800>.
48. Liang Y, Wang S, Jia H, Yao Y, Song J, Dong H, et al. Pectin functionalized metal-organic frameworks as dual-stimuli-responsive carriers to improve the pesticide targeting and reduce environmental risks. *Colloids Surf, B*. 2022;219:112796. <https://doi.org/10.1016/j.colsurfb.2022.112796>.
49. Huang G, Deng Y, Zhang Y, Feng P, Xu C, Fu L, et al. Study on long-term pest control and stability of double-layer pesticide carrier in indoor and outdoor environment. *Chem Eng J*. 2021;403:126342. <https://doi.org/10.1016/j.cej.2020.126342>.
50. Shan Y, Xu C, Zhang H, Chen H, Bilal M, Niu S, et al. Polydopamine-modified metal-organic frameworks, NH₂-Fe-MIL-101, as pH-sensitive nanocarriers for controlled pesticide release. *Nanomaterials*. 2020;10:2000. <https://doi.org/10.3390/nano10102000>.
51. Kumar R, Farda B, Mignini A, Djebaili R, Koolman L, Paul A, et al. Microbial solutions in agriculture: enhancing soil health and resilience through bio-inoculants and bioremediation. *Bacteria*. 2025;4(3):28. <https://doi.org/10.3390/bacteria4030028>.
52. Qi Y, Shang W, Wang L, Zhang J, Ngo HH, Guo W, et al. Microorganism–metal organic framework hybrids for environmental applications: interface design, assembly strategies and functional synergies. *Coord Chem Rev*. 2026;552:217529. <https://doi.org/10.1016/j.ccr.2025.217529>.
53. Li W, Chen J, Guo J, Chan KT, Liang Y, Chen M, et al. Exploring the multifaceted roles of metal-organic frameworks in ecosystem regulation. *J Mater Chem B*. 2025;13:2272–94. <https://doi.org/10.1039/D4TB01882F>.
54. Li Y, Tang S, Cong Z, Lu D, Yang Q, Chen Q, et al. Biohybrid bacterial microswimmers with metal-organic framework exoskeletons enable cytoprotection and active drug delivery in a harsh environment. *Mater Today Chem*. 2022;23:100609. <https://doi.org/10.1016/j.mtchem.2021.100609>.
55. Li M, Li L, Wang S, Sun Y, Zhang H, Li F. Development of MOF exoskeleton with robust cytoprotection on bacterial cells enables enhanced bioremediation of pesticide pollution in soil. *Pestic Biochem Physiol*. 2025;213:106534. <https://doi.org/10.1016/j.pestbp.2025.106534>.
56. Zhao Z, Liu R, Yu J, Liu Y, Li M, Yu Q. Synthetic protein-assisted Co-assembly of zeolitic imidazolate framework-8 and Novosphingobium capsulatum for enhanced saline-alkali resistance of wheat. *Molecules*. 2025;30:3669. <https://doi.org/10.3390/molecules30183669>.
57. Morales-Cámara S, Parra-Torrejón B, Rodríguez-Díéguez A, Delgado-López JM, Ramírez-Rodríguez GB, Rojas S. ZIF-8@hydroxyapatite composite as a high potential material for prolonged delivery of agrochemicals. *ACS Appl Mater Interface*. 2024;16:29305–13. <https://doi.org/10.1021/acsaami.4c06016>.
58. Mahmoud NE, Abdelhameed RM. Influence of different metal-organic frameworks on agronomic traits of faba bean plants. *BMC Plant Biol*. 2025;25:1–21. <https://doi.org/10.1186/s12870-025-07478-7>.

59. Gomathy M, Sabarinathan K, Subramanian K, Ananthi K, Kalaiyarsi V, Jeyshri M, et al. Rhizosphere: niche for microbial rejuvenation and biodegradation of pollutants. In: Microbial rejuvenation of polluted environment. Vol. 1. Springer; 2021. p. 1–22. https://doi.org/10.1007/978-981-15-7447-4_1.
60. Tian M, Guo Y, Zhou M, Yu P, Li M, Li Z, et al. Efficient preparation of high-purity sulforaphene: selective immobilization of Myrosinase by designing a concanavalin A-Loaded metal-organic framework. *J Agr Food Chem*. 2025;73:22617–25. <https://doi.org/10.1021/acs.jafc.5c06629>.
61. Farasati Far B. Antimicrobial properties of metal-organic frameworks. Logic for Metal–organic framework selection: MOFs for biomedical applications. ACS Publications; 2024. p. 147–80. <https://doi.org/10.1021/bk-2024-1463.ch007>.
62. Chakraborty S, Lynch I. Biomolecular transformations shape the environmental fate of nanoscale and emerging materials. *Acc Chem Res*. 2025;58:3316–27. <https://doi.org/10.1021/acs.accounts.5c00587>.
63. Zuhra Z, Sharafat S, Ahmad M, Ali S, Zhaowei Z. Metal-organic framework applications for microplastic remediation: exploring pathways and future potential. *J Mater Chem A*. 2025;13:9619–42. <https://doi.org/10.1039/D4TA07876D>.
64. Liang W, Wied P, Carraro F, Sumbly CJ, Nidetzky B, Tsung C-K, et al. Metal-organic framework-based enzyme biocomposites. *Chem Rev*. 2021;121:1077–129. <https://doi.org/10.1021/acs.chemrev.0c01029>.
65. Liu J, Liang J, Xue J, Liang K. Metal-organic frameworks as a versatile materials platform for unlocking new potentials in biocatalysis. *Small*. 2021;17:2100300. <https://doi.org/10.1002/sml.202100300>.
66. Wen Y, Feng M, Zhang P, Zhou H-C, Sharma VK, Ma X. Metal organic frameworks (MOFs) as photocatalysts for the degradation of agricultural pollutants in water. *ACS ES amp;T Eng*. 2021;1:804–26. <https://doi.org/10.1021/acsestengg.1c00051>.
67. Safy ME, Amin M, Haikal RR, Elshazly B, Wang J, Wang Y, et al. Probing the water stability limits and degradation pathways of metal-organic frameworks. *Chem-Eur J*. 2020;26:7109–17. <https://doi.org/10.1002/chem.202000207>.
68. Liu B, Vikrant K, Kim K-H, Kumar V, Kailasa SK. Critical role of water stability in metal-organic frameworks and advanced modification strategies for the extension of their applicability. *Environ Sci: Nano*. 2020;7:1319–47. <https://doi.org/10.1039/C9EN01321K>.
69. Chen C, Shen L, Wang B, Lu X, Raza S, Xu J, et al. Environmental applications of metal-organic framework-based three-dimensional macrostructures: a review. *Chem Soc Rev*. 2025;54:2208–45. <https://doi.org/10.1039/D4CS00435C>.
70. Wan M, Song S, Feng W, Shen H, Luo Y, Wu W, et al. Metal-organic framework (UiO-66)-based temperature-responsive pesticide delivery system for controlled release and enhanced insecticidal performance against *spodoptera frugiperda*. *ACS Appl Bio Mater*. 2022;5:4020–27. <https://doi.org/10.1021/acsabm.2c00549>.
71. Ma Y, Zhao K, Ding Y, Wu S, Liao X, Yin X, et al. A facile one-pot route to fabricate clothianidin-loaded ZIF-8 nanoparticles with biocompatibility and long-term efficacy. *Pest Manag Sci*. 2023;79:2603–10. <https://doi.org/10.1002/ps.7440>.
72. Ren M, Wang F, Xu Z, Zhang J, Wang C-C, Wang K, et al. Composite material derived from ZIF-67 and biochar promotes ozonation of 4-nitrophenol. *Chemosphere*. 2023;338:139495. <https://doi.org/10.1016/j.chemosphere.2023.139495>.
73. Chakhtouna H, Benzeid H, Zari N, Qaissa A, Bouhfid R. Microwave-assisted synthesis of MIL-53(Fe)/biochar composite from date palm for ciprofloxacin and ofloxacin antibiotics removal. *Separat Purif Technol*. 2023;308:122850. <https://doi.org/10.1016/j.seppur.2022.122850>.
74. Kong F, Lu S. Effects of microbial organic fertilizer (MOF) application on cadmium uptake of rice in acidic paddy soil: regulation of the iron oxides driven by the soil microorganisms. *Environ Pollut*. 2022;307:119447. <https://doi.org/10.1016/j.envpol.2022.119447>.
75. Bharti A, Jain U, Chauhan N. Metal-organic-frameworks: as revolutionizing material utilized for distinct applications in precision farming. *Bionanoscience*. 2024;14:1798–818. <https://doi.org/10.1007/s12668-024-01356-6>.
76. Chen H, Yang L, Zhao S, Xu H, Zhang Z. Long-term toxic effects of iron-based metal-organic framework nanopesticides on earthworm-soil microorganism interactions in the soil environment. *Sci Total Environ*. 2024;917:170146. <https://doi.org/10.1016/j.scitotenv.2024.170146>.
77. Kaur H, Kaur H, Kaur H, Srivastava S. The beneficial roles of trace and ultratrace elements in plants. *Plant Growth Regul*. 2023;100:219–36. <https://doi.org/10.1007/s10725-022-00837-6>.
78. Nikzad MS, Qiu J, Wang G, Demo NS, Liu J, Li A. Copper-and zinc-based metal-organic frameworks (MOFs) as sustainable algicides: effective inhibition of *karenia selliformis* blooms and toxin production with ecotoxicological assessment on *Artemia salina*. *Aquat Toxicol*. 2025;107689. <https://doi.org/10.1016/j.aquatox.2025.107689>.
79. Lu X, Jayakumar K, Wen Y, Hojjati-Najafabadi A, Duan X, Xu J. Recent advances in metal-organic framework (MOF)-based agricultural sensors for metal ions: a review. *Microchim Acta*. 2024;191(1):58. <https://doi.org/10.1007/s00604-023-06121-2>.
80. Rojas S, Rodríguez-Diéguez A, Horcajada P. Metal-organic frameworks in agriculture. *ACS Appl Mater Interface*. 2022;14:16983–7007. <https://doi.org/10.1021/acsami.2c00615>.
81. Asemave K, Lubem DC. Assessment of Fe-based organic framework (Fe-MOF) as nutrient-slow-releasing material. *Curr. Res. Interdiscip. Stud*. 2024;3:1–10. <https://doi.org/10.58614/cris311>.
82. Jagirani MS, Zhou W, Nazir A, Akram MY, Huo P, Yan Y. A recent advancement in food quality assessment: using MOF-based sensors: challenges and future aspects. *Crit Rev Analytical Chem*. 2025;55:581–602. <https://doi.org/10.1080/10408347.2023.2300660>.
83. Wright AM, Kapelewski MT, Marx S, Farha OK, Morris W. Transitioning metal-organic frameworks from the laboratory to market through applied research. *Nat Mater*. 2025;24:178–87. <https://doi.org/10.1038/s41563-024-01947-4>.
84. Ma S, Wang Y, Yang X, Ni B, Lü S. MOF hybrid for long-term pest management and micronutrient supply triggered with protease. *ACS Appl Mater Interface*. 2022;14:17783–93. <https://doi.org/10.1021/acsami.2c01187>.
85. Ji Y, Ma S, Lv S, Wang Y, Lu S, Liu M. Nanomaterials for targeted delivery of agrochemicals by an all-in-one combination strategy and deep learning. *ACS Appl Mater Interface*. 2021;13:43374–86. <https://doi.org/10.1021/acsami.1c11914>.
86. Bhattacharya R, Chattopadhyay MK, Datta S, Banerjee P. Metal-organic frameworks in action: a comprehensive review on fluorimetric sensing and adsorption of organophosphorus pesticides. *ChemistrySelect*. 2025;10:e01815. <https://doi.org/10.1002/slct.202501815>.
87. Han Z, Yang Y, Rushlow J, Huo J, Liu Z, Hsu Y-C, et al. Development of the design and synthesis of metal-organic frameworks (MOFs)-from large scale attempts, functional oriented modifications, to artificial intelligence (AI) predictions. *Chem Soc Rev*. 2025;54:367–95. <https://doi.org/10.1039/D4CS00432A>.
88. Chen H, Shan Y, Xu C, Bilal M, Zhao P, Cao C, et al. Multifunctional γ -cyclodextrin-based metal-organic frameworks as avermectins carriers for controlled release and enhanced acaricidal activity. *ACS Agric Sciamp; Technol*. 2023;3:190–202. <https://doi.org/10.1021/acscagtech.2c00295>.

89. Chakraborty D, Yurdusen A, Mouchaham G, Nouar F, Serre C. Large-scale production of metal-organic frameworks. *Adv Funct Mater.* 2024;34:2309089. <https://doi.org/10.1002/adfm.202309089>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.